



International trade and the risk in bilateral exchange rates[☆]

Ramin Hassan^a, Erik Loualiche^{b,*}, Alexandre R. Pecora^c, Colin Ward^b

^a Cornerstone Research, United States

^b University of Minnesota, Carlson School of Management, 321 19th Avenue South, Minneapolis, MN 55455, United States

^c Virginia Tech, Pamplin College Of Business, 880 West Campus Drive, Blacksburg, VA 24061, United States

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ABSTRACT

Exchange rate volatility falls after a trade deal, driven by a decline in the systematic component of risk. The average trade deal increases trade by 50 percent over five years, reducing systematic risk by a third of a standard deviation across countries. We examine this connection in an Armington model where the structure of trade networks determines the risk in exchange rates. We estimate our model to current data and find i) that countries at the periphery of the world trade network benefit the most from lower trade barriers and ii) that a counterfactual experiment of a trade war between the US and China shows a global increase in currency risk, with effects concentrated among peripheral countries.

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Introduction

Connecting international finance and macroeconomics rests on the dynamics of exchange rates. While our theories have long predicted currency values to be strongly affected by fundamentals like output and trade, a robust empirical link has remained remarkably elusive. The “exchange rate disconnect puzzle” summarizes a massive body of work’s struggle at bridging exchange rates and fundamentals (see for example [Meese and Rogoff \(1983\)](#) and [Engel and West \(2005\)](#)).

Recently, however, [Verdelhan \(2018\)](#) reveals the presence of a strong factor structure in bilateral exchange

rates. Refocusing away from the variation of individual exchange rates and towards understanding their systematic risk provides a new approach to reconciling the puzzle. It appeals to the wisdom of portfolio construction that dates back to at least [Black et al. \(1972\)](#) that isolates the common components of exchange rates from their idiosyncratic components.

In this paper, we present new evidence on the direct impact of international trade on currencies’ systematic risks using the surge in imports following trade agreements. In doing so, we establish not only that trade policy affects countries’ discount rates, which influence the cost of hedging and the flow of capital, but also a solid step towards unifying finance and international trade. We build a model based on these findings to evaluate different counterfactuals of trade agreements on exchange rate fluctuations.

A novelty in our work is the use of trade deals as an instrument in uncovering the effect of trade on currency risk. We begin in [Section 1](#) by confirming [Baier and Bergstrand \(2007\)](#)’s result that a typical agreement grows bilateral trade by 50 percent over the next five years. We then show that instrumented trade growth reduces the exchange rate’s systematic risk by a third of a standard deviation across countries.

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* Corresponding author.

E-mail addresses: hassa417@umn.edu (R. Hassan), eloualich@umn.edu (E. Loualiche), reggi003@umn.edu (A.R. Pecora), cward@umn.edu (C. Ward).

Our three measures of systematic risk follow recent developments in international finance. We estimate currency betas and record the *R*-squareds from regressions of changes in a bilateral exchange rate on a country's base factor. Akin to market portfolios, base factors average out all foreign country-specific shocks, leaving exposure to only domestic shocks and global common factors. To these we add a new measure of unshared risk that divides a bilateral pair's exchange rate variance (how much risk is not shared) by the sum of the pair's base factor variances (how much risk there is to share). It has an economic interpretation close to (one minus) the risk-sharing index of [Brandt et al. \(2006\)](#), but is constructed using only currency data. For all measures, an exchange rate's systematic risk falls when two countries become “closer”, in the sense that their pricing kernels now load more identically on a common set of factors. More similar exposure translates all else equal into more stable exchange rates.

In our empirical analysis we run a multi-year difference specification in panel data to control for self-selection into trade deals following [Baier et al. \(2014\)](#). We do this by including pre-trends and country-year and country-pair fixed effects. The country-year effects capture changes in importer and exporter characteristics such as multilateral resistance terms, and in this difference specification the country-pair effects control for unobservable pair-specific changes that could have occurred ([Trefler, 2004](#)). Our identifying assumption is that trade deals only impact currency risk through their effect on trade.

To examine the robustness of our results, we split our deals into two dimensions of types: distinguishing multilateral and bilateral deals, including those where neither country coincidentally changes their currency anchor ([Ilzetzki et al., 2019](#)); and separating out free trade agreements, customs unions, and preferential tariff agreements. We find similar results to our headline estimates, especially among free trade agreements, so we are confident in the validity of our exclusion restriction.

Furthermore, some of our exchange rate risk measures are not devoid of country-specific idiosyncratic risk. In consequence, we employ a principal component analysis on the cross-section of the base factors to construct new risk measures that plausibly reflect only systematic risk. We find that this purely systematic component declines in response to a trade deal and that, as an important placebo test, the residual idiosyncratic component does not.

We then build a structural model to uncover the theoretical underpinnings of how trade frictions impact the structure of exchange rate risk. We think of trade deals as representing a change in trade costs, which represent a broad notion of barriers—physical, institutional, cultural, informational, tariff, and non-tariff, like domestic shipping regulations—that impede the transportation of goods and thus risk sharing. Accordingly, in [Section 2](#) we formulate an international economy of trade in goods hindered by both trade costs and asset market frictions. The model, which borrows from the theoretical foundations of the gravity equation, endogenizes the currency factor structure from granular origins ([Gabaix, 2011](#)).

Key terms in gravity models are those of multilateral resistance, equilibrium constructs that influence bilateral

trade and that depend on the network of *all* countries' trade costs. Since [Anderson and van Wincoop \(2003\)](#) it has been known that accounting for these terms is crucial for the estimation of trade costs which otherwise would be biased from omitted variables. Since trade costs are a first-order determinant of a country's exposure to global risk, it is plausible that this bias spills into exchange rates. We therefore structurally estimate trade costs in [Section 3](#) before evaluating the model.

We first show that our estimated gravity model predicts core countries are larger, face lower resistances, and are more exposed to global shocks relative to peripheral countries. When an adverse global shock occurs, core countries' currencies appreciate relative to peripheral countries, causing investors in the core to perceive peripheral currencies as risky *ex ante*, consistent with patterns in the data. We then examine counterfactual experiments on trade policy's role in determining the covariation of exchange rates in [Section 4](#), as recent work in international macroeconomics has highlighted the importance of exchange rates in affecting firm policies.

In our first counterfactual experiment, we validate our empirical setting. We implement a change in trade costs to replicate the import growth observed empirically in the first stage and to compare the model's predictions for the second stage with the data. It allows us to see how global shock exposures change as a function of a country's characteristics in response to a trade agreement, and thus where the gains from trade concentrate. We find that peripheral countries, which are often poorer, are found to benefit most from a reduction in trade costs, consistent with our empirical findings. Intuitively, trade deals that substantially weaken a peripheral's multilateral resistance are effective in bringing it into the trade network's core. Because they faced the highest barriers *ex ante*, peripheral countries experience the greatest relative reductions *ex post*.

Next, we study the impact of a trade war between two major economies, the United States and China, on the risk exposures of other countries. Our counterfactual shows that these two countries' sizes largely insulate them from adverse consequences, while peripheral countries bear collateral damage and the brunt of the war—peripherals' currency volatilities rise, chiefly attributable to a growth in their systematic risk exposure. Because of the trade war of a large two, risk-sharing across all has worsened. Overall, our paper underscores that trade deals are not just about trade.

Related Literature. Our work continues to uncover the fundamental drivers of currency prices. The empirical literature on the exchange rate disconnect puzzle is quite extensive and began with [Meese and Rogoff \(1983\)](#) who showed that exchange rate forecasts based on a random walk became less accurate by conditioning on macroeconomic variables.¹ A fresh perspective on the puzzle comes from [Engel et al. \(2015\)](#) and [Verdelhan \(2018\)](#) who demon-

¹ See [Frankel and Rose \(1995\)](#) and [Cheung et al. \(2005\)](#) for surveys. More recently, [Lilley et al. \(2019\)](#) look at how fundamentals and exchange rates have become more reconnected from 2007 onward.

strate that a large amount of variation in bilateral exchange rates is explained by common factors.

Building on this work, [Lustig and Richmond \(2020\)](#) relate currency exposures to gravity-type measures of distance, whether it be cultural, institutional, or simply geographical. [Jiang and Richmond \(2020\)](#) parameterize a network of international trade using import shares to develop a measure of network closeness; they show that closer countries have more correlated consumption growth rates, stock returns, and exchange rate movements. Nevertheless, these two papers center on time-invariant measures and parameterizations. An advantage of our focus on trade deals is that we do not rely on the structural relation between geography and trade flows. Rather, we explore variation in the data from the time series that maps directly into our counterfactual exercises. Our trade network and therefore exchange rate factor structure, moreover, are endogenous and change over time in response to trade deals.

A subsequent branch of research in international economics focuses on the weak correlations between exchange rates and macroeconomic quantities that are surprisingly different from those predicted by theory ([Backus et al. \(1992\)](#), [Backus and Smith \(1993\)](#), [Evans and Lewis \(1995\)](#), and [Lewis \(1996\)](#)). [Obstfeld and Rogoff \(2000\)](#) and [Anderson and van Wincoop \(2004\)](#) argue that trade costs are key to resolving several major puzzles in international economics and they have been extensively studied in this literature. These costs, however, have played only a modest role in international finance: see [Dumas \(1992\)](#), [Hollifield and Uppal \(1997\)](#), [Verdelhan \(2010\)](#), [Ready et al. \(2017a\)](#), [Ready et al. \(2017b\)](#), [Maggiore \(2017\)](#), and [Barrot et al. \(2019\)](#), although these papers use two-country models that are unsuitable to study the cross-section of bilateral exchange rates. We instead specify a multi-country setup that is closer to [Fitzgerald \(2012\)](#)'s, but in contrast to her work we look at our model's implications for the exchange rate factor structure.²

Our interest in systematic risk places the paper into the large literature on sources of foreign exchange return premia.³ [Hassan \(2013\)](#) shows that the size of a country is an important determinant of the cross-sectional variation in currency returns, simply because larger countries' bonds better insure against shocks that affect a larger fraction of the world economy. [Richmond \(2019\)](#) focuses on the propagation of shocks through the hubs of the global trade network, identifying countries like the Netherlands and Singapore as more exposed to global risk. Like these two papers, our model only quantitatively generates the quantity of risk and does not jointly account for its price, for example by including disasters or long-run risk. Unlike them, size and network centrality are endogenous to our model's structure of trade costs. We therefore can, and do, evaluate

how these characteristics change over time in response to trade policy.

1. The role of trade in exchange rate variation

We first review the measurement of the systematic risk of bilateral exchange rates. We then discuss data sources before turning to our empirical results.

1.1. Measurement of currency risk

To motivate our measures we follow the literature ([Verdelhan \(2018\)](#) for example) and start by specifying innovations to log stochastic discount factors (SDFs) in complete markets. From these we construct exchange rate dynamics and decompose them into measures of risk. Alternatively, in [Appendix A.1](#) we show how we can derive identical measures of risk by specifying dynamic processes for exchange rates directly without relying on the assumption of complete markets.

1.1.1. Dynamics of exchange rates

Country i 's SDF follows a two-factor process

$$-\Delta \mathbf{E}_t[m_{i,t+1}] = \delta_i u_{i,t+1} + \varphi_i u_{g,t+1}. \quad (1)$$

For compactness we define the innovations operator, $\Delta \mathbf{E}_t[x_{t+1}] \equiv x_{t+1} - \mathbf{E}_t[x_{t+1}]$, as we are only interested in second moments. While our exogenous shocks here, u_i and u_g , do not translate directly into real shocks, in the model of [Section 2](#) we show how they map into country-level supply shocks and how gravity granularly creates global shocks.⁴ For now, we interpret u_i as a country-specific shock that is uncorrelated across countries and u_g as a global aggregate shock that is orthogonal to all u_i ; the respective (positive) exposures to these shocks are δ_i and φ_i . The shocks' means are zero and their distributions are otherwise unrestricted, but for illustration we set all variances equal to σ^2 for what follows.

Under complete markets, the innovation to a log return of a spot exchange rate is the difference between country j 's and i 's SDF innovations. Given its specification we have the following innovation dynamics for the exchange rate's log return:

$$\begin{aligned} \Delta \mathbf{E}_t[\Delta s_{ij,t+1}] &= \Delta \mathbf{E}_t[m_{j,t+1}] - \Delta \mathbf{E}_t[m_{i,t+1}] \\ &= \delta_i u_{i,t+1} - \delta_j u_{j,t+1} + (\varphi_i - \varphi_j) u_{g,t+1}. \end{aligned} \quad (2)$$

The exchange rate is the price of foreign currency i per unit of the domestic currency j . To fix ideas, label the United States as the domestic currency, though in our empirics we consider different domestic countries. When an adverse shock originates from the US ($u_{j,t+1} < 0$), the dollar appreciates. Furthermore, if the US has a relatively large global exposure, $\varphi_j > \varphi_i$, then coincident with an adverse global shock, the dollar also appreciates. These phenomena are consistent with a narrative of the US dollar being a safe

² Few studies in international finance use multi-country models, though none specify trade costs: [Heyerdahl-Larsen \(2014\)](#), [Mueller et al. \(2017\)](#), [Colacito et al. \(2018b\)](#).

³ A partial list is [Lustig and Verdelhan \(2007\)](#), [Colacito and Croce \(2011\)](#), [Lustig et al. \(2011\)](#), [Colacito and Croce \(2013\)](#), [Gabaix and Maggiori \(2015\)](#), [Lettau et al. \(2014\)](#), [Corte et al. \(2016\)](#), [Stathopoulos \(2017\)](#), [Colacito et al. \(2018a\)](#), [Chernov et al. \(2020\)](#).

⁴ [Burnside and Graveline \(2020\)](#) argue structural assumptions about frictions in goods markets and preferences are needed to truly interpret risk-sharing and variation in exchange rates.

haven. The notion of closeness among country pairs we refer to can be easily seen: as φ_i becomes closer to φ_j , the exchange rate's exposure to global shocks goes to zero.

Country j 's base factor is defined as the average appreciation of all currencies with respect to it: $\Delta base_{j,t+1} \equiv \frac{1}{N} \sum_{i=1}^N \Delta s_{ij,t+1}$. Lustig and Richmond (2020) show it correlates highly with each country's level factor, the first principal component extracted from all bilateral exchange rates with respect to the base country. In this literature the large N limit is often taken for clarity, and following this approach innovations to j 's base factor become

$$\Delta E_t[\Delta base_{j,t+1}] = -\delta_j u_{j,t+1} + (\bar{\varphi} - \varphi_j) u_{g,t+1}, \quad (3)$$

where we denote the average with $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$, with some abuse of notation. In this limit, other countries' idiosyncratic shocks average out and we find that j 's base factor loads only on its own shock u_j and the global shock u_g . The latter depends on the difference between its own SDF loading and the average foreign currency loading on the global shock. If all countries were to share similar φ , then all SDFs would comove with the global shock identically and j 's base factor's loading on the shock would be close to zero. Thus, base factors help isolate country-level variation in global shock exposure.

1.1.2. Measuring the systematic variation in bilateral exchange rates

To recover the quantity of systematic currency risk in i taken on by a currency investor from j , we regress the pair's exchange rate return on j 's base factor. The slope coefficient recovers the rate's common exposure to global risk⁵

$$\Delta s_{ij,t+1} = a_{ij} + \beta_{ij} \Delta base_{j,t+1} + \varepsilon_{ij,t+1}. \quad (4)$$

We consider four measures of currency risk: volatility σ , the slope coefficient β , the regression's R^2 , and unshared risk, ρ . As N gets large, they become

$$\sigma_{ij}^2 = \text{Var}(\Delta s_{ij,t+1}) = \sigma^2(\delta_i^2 + \delta_j^2 + (\varphi_i - \varphi_j)^2) \quad (5)$$

$$\beta_{ij} = \frac{\text{Cov}(\Delta s_{ij,t+1}, \Delta base_{j,t+1})}{\text{Var}(\Delta base_{j,t+1})} = \frac{\delta_j^2 + (\bar{\varphi} - \varphi_j)(\varphi_i - \varphi_j)}{\delta_j^2 + (\bar{\varphi} - \varphi_j)^2} \quad (6)$$

$$R_{ij}^2 = \beta_{ij}^2 \frac{\text{Var}(\Delta base_{j,t+1})}{\text{Var}(\Delta s_{ij,t+1})} = \beta_{ij}^2 \times \frac{\delta_j^2 + (\bar{\varphi} - \varphi_j)^2}{\delta_i^2 + \delta_j^2 + (\varphi_i - \varphi_j)^2} \quad (7)$$

$$\begin{aligned} \rho_{ij} &= \frac{\text{Var}(\Delta s_{ij,t+1})}{\text{Var}(\Delta base_{i,t+1}) + \text{Var}(\Delta base_{j,t+1})} \\ &= \frac{\delta_i^2 + \delta_j^2 + (\varphi_i - \varphi_j)^2}{\delta_i^2 + (\bar{\varphi} - \varphi_i)^2 + \delta_j^2 + (\bar{\varphi} - \varphi_j)^2}. \end{aligned} \quad (8)$$

⁵ In the case where the domestic country is the US, Verdelhan (2018) shows that the slope coefficient is the loading on the dollar factor, which explains a large fraction of the variation in bilateral dollar exchange rates.

Writing about these in turn, it is clear that volatility is the least preferred measure of systematic risk as its estimate is contaminated by country-specific risk and σ^2 , which could be time-varying. The slope coefficient is a ratio that naturally improves on σ_{ij} by down-weighting variation due to country-specific risk. Analogous to a stock market beta, β_{ij} records the incremental systematic risk that an investor in j takes on when investing in i 's currency. The coefficient can be increasing or decreasing in φ_i , however, depending on φ_j 's location relative to $\bar{\varphi}$ and φ_i . For this reason, R -squared is a betterment on β_{ij} . Squaring both $(\varphi_i - \varphi_j)$ and $(\bar{\varphi} - \varphi_j)$ produces monotonicity of R_{ij}^2 with respect to the distance of i from j . It measures the fraction of variance of i 's currency due to systematic risk from j 's perspective.

To these measures we add unshared risk that does not require a regression framework. As in R -squared, the squaring of all differences yields monotonicity. It has an economic definition very close to (one minus) the risk-sharing index of Brandt et al. (2006): this can be seen by using (1) to rewrite (8) as

$$\rho_{ij} = \frac{\text{Var}(m_{j,t+1} - m_{i,t+1})}{\text{Var}(m_{i,t+1}) + \text{Var}(m_{j,t+1}) + 2\bar{\varphi}(\bar{\varphi} - \varphi_i - \varphi_j)}, \quad (9)$$

where the only small difference is in the denominator, $2\bar{\varphi}(\bar{\varphi} - \varphi_i - \varphi_j)$.⁶ It therefore inherits the intuition that countries that better share risk internationally have lower unshared risk. A key difference is that our risk-sharing index only requires currency data and bypasses the difficulty in estimating the mean return on the stock market needed to implement the Hansen-Jagannathan bound.

Altogether, we prefer R -squared and unshared risk to the other two risk measures, but we report all as they each possess a distinct economic meaning. All four are expected to decrease as countries become closer.

1.2. Data

Our sources draw from databases on exchange rates, international trade, macroeconomic aggregates, and trade deals. We discuss these in turn before describing our construction of the risk measures.

1.2.1. Exchange rates and pegs

We use exchange rate data from Bloomberg that span 1973 to 2019 at the daily frequency, as we focus on the accurate estimation of second moments. All exchange rates are relative only to the US dollar and we triangulate to recover all bilateral pairs. Our 45 countries are the developed and emerging markets listed by Morgan Stanley Capital International as of June 2020. We follow Lustig and Richmond (2020) and omit currencies after joining the Euro, include the Euro throughout when constructing base factor loadings, and drop the Euro countries in regressions due to lack of gravity data. We identify pegged currencies using Harms and Knaze (2021) bilateral classification of the Ilzetzi et al. (2017) data. Specifically, countries that have

⁶ Our structurally estimated model recovers this empirically unobservable term to be near -0.035, much smaller than $2 \text{Var}(m)$ that stock market data tells us is over 0.50 (see Brandt et al. (2006)' footnote 2).

a de facto peg or stronger (indices one through four) are classified as having pegged currencies.

1.2.2. Fundamentals and trade deals

Trade flows are from the IMF's Direction of Trade Statistics database from 1973 to 2019. The IMF trade data do not include services, only the value of merchandise exports and imports. We obtain data on output (GDP), inflation (CPI), and consumption from the World Bank's Development Indicators and on gravity variables from CEPII's Research and Expertise on the World Economy (Head and Mayer (2014)).

The data on trade agreements comes from the database on Economic Integration Agreements (EIA) constructed by Baier and Bergstrand (2007).⁷ The data cover annually the economic integration level for bilateral country pairings since 1950. Every year each country pair is assigned one of seven distinct levels of integration. The levels in order of increasing intensity are

1. No Agreement
2. Non-Reciprocal Preferential Trade Agreement (preferential terms and customs concessions given by developed countries to developing ones, such as the US's Generalized System of Preferences)
3. Preferential Trade Agreement (preferential terms to members of the agreement versus non-members)
4. Free Trade Agreement (trade barriers substantially eliminated among members, but each member can treat non-members differently)
5. Customs Union (same as free trade arrangement, but treat non-members the same)
6. Common Market (same as customs union, but also includes free movement of labor and capital) and
7. Economic Union (same as common market, but also monetary and fiscal policy coordination)

Because of the small number of deeper EIAs, we combine customs unions, common markets, and economic unions into one agreement, CUCMECU, leaving our other four classifications as no agreement, one-way and two-way preferential trade agreement (OWPTA and TWPTA), and free trade agreements (FTA).

1.2.3. Computing risk and summary statistics

We use equation (4), $\Delta s_{ij,t+1} = a_{ij} + \beta_{ij} \Delta base_{j,t+1} + \varepsilon_{ij,t+1}$, to estimate β_{ij} and R^2_{ij} . Here, $\Delta s_{ij,t+1}$ is the daily change in the value of the foreign currency quoted in domestic currency j and the base factor is the (equally-weighted) average of the foreign currencies quoted in the domestic currency.

Given our interest in the change of risk around trade deals, we estimate rolling regressions for each day using a 1,800-day window (corresponding to approximately 5 years) and restrict country-pairs to where there are at least 100 days of data. We compute daily measures of volatility σ_{ij} and unshared risk ρ_{ij} with the same rolling window. We then aggregate all measures by averaging over all

daily estimates within a year. In the end, we are left with a panel of country-pairs and yearly estimates of currency risk. Because volatility is a daily estimate, for comparison we annualize it with the square root rule.

We present summary statistics of bilateral exchange rate and bilateral trade in Table 1. All moments come from annual country-pair observations following the aggregation described above. The average change in currency prices over a year is effectively zero (not reported) with an annualized volatility of 14.4 percent and a median of 11.9 percent. By construction, the average base factor loading β is close to one. The loading varies widely across country-pairs as can be seen from its high standard deviation. The average R -squared of the estimated regressions means that approximately half of the variation in bilateral exchange rates is explained by systematic risk.

We split the sample between developed and emerging markets and present summary statistics by group in Appendix Table A.1. Emerging market's average R -squared is relatively higher.

In the lower panel of Table 1, we present statistics for bilateral imports, our measure of trade. The average across EIA variables tells the percentage of yearly country-pairs in our sample that had an agreement. For example, we observe 8.8 percent of observations under a CUCMECU agreement. Unshared risk has an average of over 90 percent and is possibly quite high because of around three-fifths of our panel reporting no trade agreement at some point in time, highlighting the substantial distortions to international risk sharing that trade costs potentially generate. As time passes, however, every country eventually completes one trade deal with another.

1.3. Trade deals and trade

We first confirm the results in Baier and Bergstrand (2007) and show bilateral trade flows are bolstered after a trade deal is signed. Precisely, we estimate

$$\Delta_{t,t+h} \log \text{Imports}_{ij,t} = \beta \text{Trade Deal}_{ij,t} + \gamma X_{ij,t} + \varepsilon_{ij,t}. \quad (10)$$

and include in our controls $X_{ij,t}$ fixed effects for year of the deal and, importantly, the country-pair to absorb unobserved pair-specific characteristics that could influence the likelihood of a trade deal. These regressions' results are tabulated in Table 2 for a window of five years (columns (1) and (2)) and ten years (columns (3) and (4)) before and after the trade deal. The estimated coefficient β shows that trade grows by 50 percent within five years after the deal and that the effect continues to cumulate for another five, reaching nearly 70 percent ten years after the deal.

This specification has a low amount of variation, however, as it only has a before and after for every deal—the country-pair fixed effect explains virtually all variation. More concerning is that trade and the likelihood of a deal are slow-moving variables that could be influenced by macroeconomic conditions, so our estimates could be correlated with these unobservable factors.

⁷ The dataset is available on Jeffrey Bergstrand's website at <https://sites.nd.edu/jeffrey-bergstrand/database-on-economic-integration-agreements/> (last accessed on August 23, 2021).

Table 1
Summary Statistics Across Countries

	Observations	Mean	Median	Std. Dev.
<i>Risk Measures</i>				
Base Loadings, β	55,780	1.012	0.995	0.745
Share of Systematic Risk, R^2	55,742	0.497	0.498	0.307
Volatility, σ	55,780	0.144	0.119	0.123
Unshared Risk, ρ	55,780	0.911	0.965	0.309
<i>Trade Measures</i>				
Imports (billions of USD)	93,420	6.153	0.606	28.22
Imports, five-year difference (billions of USD)	81,676	1.648	0.132	9.366
Imports ten-year difference (billions of USD)	71,178	3.675	0.389	16.57
Imports, as a share of GDP	77,648	0.026	0.006	0.085
CUMECU	112,800	0.088	0.000	0.283
FTA	112,800	0.114	0.000	0.317
TWPTA	112,800	0.035	0.000	0.185
OWPTA	112,800	0.154	0.000	0.361

This table summarizes statistics of nominal exchange rates and trade at the country level.

Base loadings and R -squareds are estimated from the regression in (4).

The import shares are estimated from IMF and World Bank data over the period 1973–2019.

Trade deals are divided in four different levels of economic integration as in [Baier and Bergstrand \(2007\)](#): customs unions, common markets, and economic unions (CUMECU), free trade agreement (FTA), one-way and two-way preferential trade agreement (OWPTA and TWPTA).

Table 2
Change in Imports with Trade Deals

	(1) 5Y window	(2) 5Y window	(3) 10Y window	(4) 10Y window
After Trade Deal	0.513*** (0.015)	0.492*** (0.015)	0.685*** (0.018)	0.656*** (0.018)
Distance	-0.000 (0.000)		-0.000** (0.000)	
Common Border	1.405*** (0.238)		1.422*** (0.235)	
Common Language	0.345 (0.251)		0.315 (0.245)	
Colonial Link	0.014 (0.362)		0.007 (0.316)	
Common Legal	0.560*** (0.154)		0.583*** (0.150)	
Constant	20.181*** (0.149)	20.367*** (0.007)	20.359*** (0.147)	20.430*** (0.009)
Observations	1468	1540	1504	1540
r2	0.341	0.990	0.356	0.985
F	210.089	1079.722	267.541	1366.202

This table reports the regression result of the change in log imports of base from foreign country around a trade deal over a five-year window in columns (1) and (2) and 10-year window in columns (3) and (4). The coefficient for “After Trade Deal” is interpreted in log changes from the average log value as expressed in the constant.

Trade deals are defined as a change in any kind of bilateral agreement to a stronger FTA or CUMECU deal (see [Section 1.2](#)).

Columns (1) and (3) controls for gravity variables from CEPII, columns (2) and (4) control for country-pair fixed effects. All regressions have a year-of-trade-deal fixed effect.

The sample period is 1973–2019 for 45 countries.

Standard errors are clustered at base-foreign country level. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively

To alleviate these problems, we follow [Baier et al. \(2014\)](#) and estimate the following model

$$\Delta_{t,t+h} \log \text{Imports}_{ijt} = \beta \Delta_{t,t-h} \text{Trade Deal}_{ijt} + \gamma X_{ijt} + \varepsilon_{ijt,t+h} \quad (11)$$

where we separately examine the role of distinct trade deals—CUMECU, FTA, TWPTA, and OWPTA—and where our set of controls X_{ijt} include an additional five-year lag in imports for pre-trends, country-year fixed effects for country-specific time-variation in macroeconomic conditions, and either gravity variables or country-pair fixed effects for unobservable pair-specific changes that could

have occurred, like a technological reduction in trade costs ([Trefler, 2004](#)).

As we see in [Table 3](#), only CUMECUs and FTAs have a significant impact on bilateral trade. Imports increased by 17 percent approximately over five years following CUMECUs and by around 14 percent following FTAs with no detectable effects for other forms of trade agreements (columns 2 and 3).

1.4. Trade and exchange rate risk

After establishing that trade deals influence trade, we now turn to our formal analysis of the effect of trade on

Table 3
First Stage: Trade and Trade Deals

	$\Delta_{t+5,t} \log \text{Import}$			
	(1)	(2)	(3)	(4)
$\Delta_{t,t-5} \text{CUCMECU}$	0.218*** (0.035)	0.189*** (0.039)	0.172*** (0.034)	0.156*** (0.033)
$\Delta_{t,t-5} \text{FTA}$	0.140*** (0.023)	0.163*** (0.026)	0.148*** (0.023)	0.155*** (0.022)
$\Delta_{t,t-5} \text{TWPTA}$	0.054 (0.034)	0.020 (0.038)		
$\Delta_{t,t-5} \text{OWPTA}$	0.057 (0.035)	0.044 (0.038)		
$\Delta_{t-5,t} \log \text{Import}$	-0.218*** (0.029)	-0.247*** (0.027)	-0.247*** (0.027)	-0.250*** (0.028)
Gravity Controls	Y	–	–	–
Country pair f.e.	N	Y	Y	Y
Country-year f.e.	Y	Y	Y	Y
Exchange rate regime	All	All	All	No Pegs
Observations	69,940	69,940	69,940	66,427
R-squared	0.315	0.378	0.378	0.379

This table shows the estimates of a first stage OLS regression of log imports on trade agreement dummies.

All regressions control for country-time fixed effects. Column (1) controls for gravity variables. In columns (2) through (4), the gravity variables are replaced by country-pair fixed effects. Column (4) also removes pegged currencies.

Standard errors are clustered at base-foreign country level. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

exchange rate risk based on two regressions: staggered difference-in-difference and two-stage least squares.

1.4.1. Staggered difference-in-difference regression

We first consider the temporal change of trade and exchange rate risk around trade deals. Our event study takes the form of a staggered difference-in-difference regression:

$$y_{ij,t} = \alpha + \sum_{s \neq t^*} \beta_s \text{Trade Deal}_{ij} \times \text{Year}_s + \gamma X_{ij,t} + \epsilon_{ij,t}. \quad (12)$$

The trade deal dummy equals one if countries i and j share either a FTA or a CUCMECU and is zero otherwise. The set of controls $X_{ij,t}$ includes gravity variables, year fixed effects, and the deal dummy. The β_s coefficients capture the effect of a trade deal on $y_{ij,t}$ for every year s before and after the year of the deal, t^* , when strengthening of any bilateral agreement to a FTA or a CUCMECU occurs. Note that we do not include $s = t^*$ in the interaction, so the coefficients β_s measure the deviation in $y_{ij,t}$ from the time of the deal.

In column (1) of Table 4 we reconfirm the impact of trade deals on trade and plot the trajectory of bilateral trade around a deal in Fig. 1. We only include FTA and CUCMECU trade agreements and find trade is bolstered by 18 percent over five years and more than 30 percent over ten.

Note that some coefficient estimates for log imports for years prior to the event are negative and statistically different from zero. This suggests some degree of anticipation, common for long trade negotiations. There is a lower level of trade until one year before the event, rather than a strong pre-trend increase.

Turning now to our first set of results on the impact of trade on currency risk, we report the estimates in columns (2) through (5) of Table 4 and plot the time paths of all risk measures in Fig. 2. All four measures display a marked

decrease that generally become statistically significant after five years. Most trade agreements are “phased-in” over five years, so it is not surprising effects materialize after a lag. The effects persist up to ten years out, spanning from a 1.2 percentage points drop in volatility to a 0.21 reduction in beta. All are economically significant.

As a robustness check, we include bilateral fixed effects in (12), strictly isolating the time-series variation across each country pair. In Appendix Table A.3, we show that the effects remain statistically significant and are comparable to our headline numbers, although the magnitudes are slightly smaller.

Our staggered design in (12) differs from the standard difference-in-difference regression as our individual units (country pairs) are not treated simultaneously but throughout the sample period. Staggering can introduce a discrepancy between the estimated coefficients and the actual average treatment effect (see Callaway and Sant’Anna (2019) and Goodman-Bacon (2020) for details). We use the methodology introduced in Callaway and Sant’Anna (2019) and report results using their estimator in Appendix Table A.4. We draw similar conclusions to our baseline estimates.

1.4.2. Two-stage least squares regression

We now move on to our alternative approach, the second stage of our two-stage least squares regression. We use the change in bilateral trade around trade deals shown in Table 3 to instrument the effect of trade growth on bilateral exchange rate risk. Our second-stage takes the form

$$\Delta_{t,t+h} y_{ij} = \alpha + \beta \Delta_{t,t-5} \log \widehat{\text{Imports}}_{ij,t} + \gamma X_{ij,t} + u_{ij,t}, \quad (13)$$

where $\Delta_{t,t-h} \log \widehat{\text{Imports}}_{ij,t}$ had been estimated in the first stage (11). We include domestic and foreign country-year fixed effects to rule out differential trends across countries

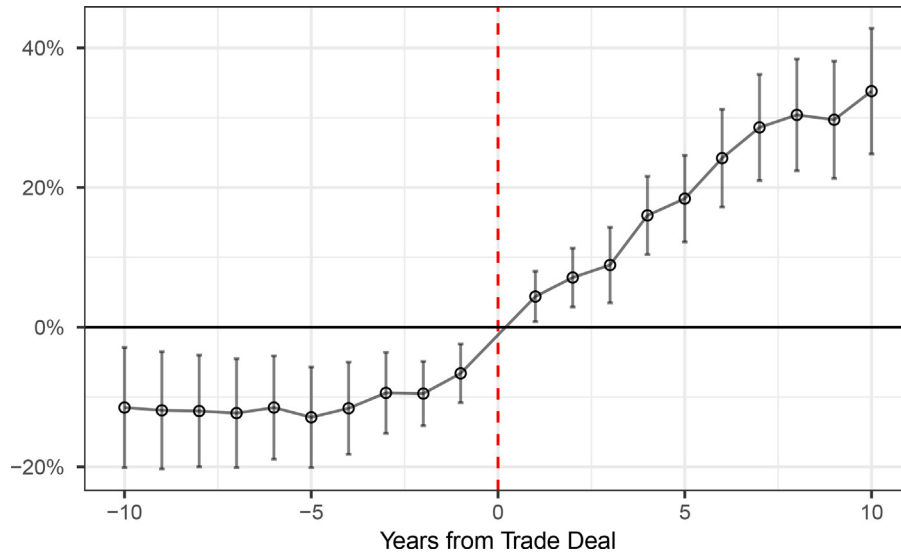


Fig. 1. Change in Log Imports Around Trade Agreements. This figure plots the estimates of our staggered difference-in-difference regression in (12) with trade as the dependent variable. We compare for each year around a trade deal country-pairs with and without an agreement. The regression includes domestic and foreign country-year fixed effects and the standard gravity controls described in Table 2. Our trade deal variable equals one if countries have a FTA or a CUCMECU and is zero otherwise.

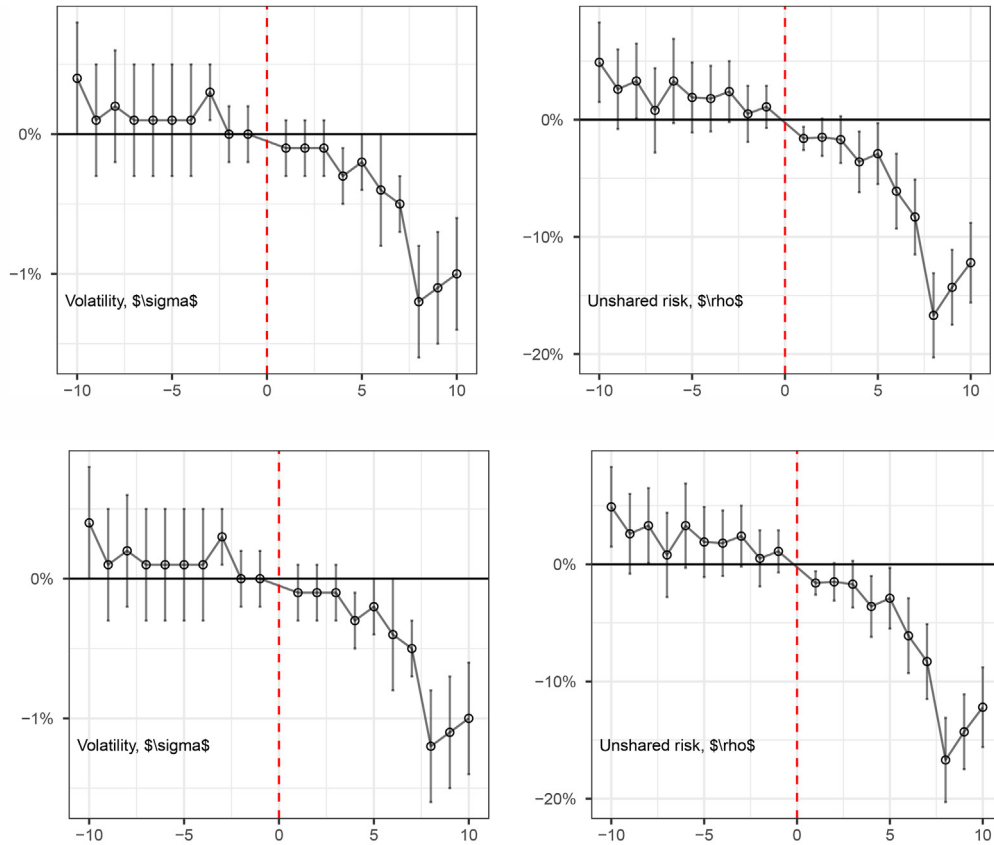


Fig. 2. Change in Currency Risk Around Trade Agreements. This figure plots the estimates of our staggered difference-in-difference regression in (12) with volatility, unshared risk, R-squared, or base loading as the dependent variable. We compare for each year around a trade deal country-pairs with and without an agreement. The regression includes domestic and foreign country-year fixed effects and the standard gravity controls described in Table 2. Our trade deal variable equals one if countries have a FTA or a CUCMECU and is zero otherwise.

Table 4
Time Effects of Trade Deal Change. Staggered Difference-in-Difference Regression.

Timing	Log Import		Volatility, σ		Unshared Risk, ρ		R-squared		Base Loadings, β	
	β	s.e.	β	s.e.	β	s.e.	β	s.e.	β	s.e.
-10	-0.115***	(0.043)	0.004**	(0.002)	0.049***	(0.017)	0.028**	(0.013)	0.130**	(0.059)
-9	-0.119***	(0.042)	0.001	(0.002)	0.026	(0.017)	0.017	(0.014)	0.091*	(0.050)
-8	-0.120***	(0.040)	0.002	(0.002)	0.033**	(0.016)	0.018	(0.013)	0.085**	(0.042)
-7	-0.123***	(0.039)	0.001	(0.002)	0.008	(0.018)	0.003	(0.015)	0.062	(0.038)
-6	-0.115***	(0.037)	0.001	(0.002)	0.033*	(0.018)	0.020	(0.016)	0.077**	(0.035)
-5	-0.129***	(0.036)	0.001	(0.002)	0.019	(0.015)	0.011	(0.013)	0.062*	(0.034)
-4	-0.116***	(0.033)	0.001	(0.002)	0.018	(0.014)	0.010	(0.012)	0.053**	(0.026)
-3	-0.094***	(0.029)	0.003**	(0.001)	0.024*	(0.013)	0.013	(0.011)	0.062***	(0.023)
-2	-0.095***	(0.023)	0.000	(0.001)	0.005	(0.012)	0.005	(0.010)	0.036*	(0.019)
-1	-0.066***	(0.021)	0.000	(0.001)	0.011	(0.009)	0.008	(0.008)	0.021	(0.013)
1	0.044**	(0.018)	-0.001	(0.001)	-0.016***	(0.005)	-0.007	(0.005)	-0.010	(0.012)
2	0.071***	(0.021)	-0.001	(0.001)	-0.015*	(0.008)	-0.006	(0.007)	-0.016	(0.013)
3	0.089***	(0.027)	-0.001	(0.001)	-0.017	(0.010)	-0.009	(0.009)	-0.031**	(0.014)
4	0.160***	(0.028)	-0.003**	(0.001)	-0.036***	(0.013)	-0.019*	(0.011)	-0.056***	(0.017)
5	0.184***	(0.031)	-0.002	(0.001)	-0.029**	(0.013)	-0.015	(0.010)	-0.042**	(0.017)
6	0.242***	(0.035)	-0.004***	(0.002)	-0.061***	(0.016)	-0.034***	(0.013)	-0.077***	(0.021)
7	0.286***	(0.038)	-0.005***	(0.001)	-0.083***	(0.016)	-0.045***	(0.012)	-0.108***	(0.021)
8	0.304***	(0.040)	-0.012***	(0.002)	-0.167***	(0.018)	-0.089***	(0.013)	-0.208***	(0.024)
9	0.297***	(0.042)	-0.011***	(0.002)	-0.143***	(0.016)	-0.077***	(0.012)	-0.200***	(0.032)
10	0.338***	(0.045)	-0.010***	(0.002)	-0.122***	(0.017)	-0.064***	(0.012)	-0.178***	(0.033)
Observations	84134		49120		49120		49082		49120	
R-squared	0.858		0.961		0.299		0.655		0.616	

This table reports coefficients from a staggered difference-in-difference regression as in (12). The deal dummy equals one if countries i and j share either an FTA or a CUCMECU and zero otherwise. The set of control variables X_{ijt} includes gravity variables, time fixed effects, and the deal dummy. All time coefficients are deviations in y_{ijt} from the exact year of the trade deal change.

driving estimates. We consider two different specifications, one controlling for gravity variables and other replacing them with a country-pair fixed effect. Our exclusion restriction assumes that trade deals only affect currency variation through their effect on trade. We present the results for the four measures of exchange rate risk across specifications in Tables 5 and 6.

In Table 5, we find that volatility and unshared risk shrink in response to a growth in bilateral trade. Given a 50 percent increase in imports, volatility falls by 0.9 (0.018×0.5 , column 1) to 1.9 (0.037×0.5 , column 2), percentage points over five years. Since average volatility is 14.4 percent, this translates to a drop of 6 to 13 percent. Unshared risk's reduction between 14 and 22 percent is more drastic, being within one- to two-thirds of its standard deviation of 30.9 percent. Both effects persist at the ten-year horizon, even when excluding pegged currencies.

We report the results using the common factor regression in (4) in Table 6. The 50 percent growth in imports yields a fall in β between 0.13 to 0.20 at the five-year horizon, over a fifth of its standard deviation. Trade reduces systematic risk as R-squared falls by approximately 7 to 11 percent at the five-year horizon or up to a third of its standard deviation across countries.

The evidence presented in these tables stand in stark contrast to a simple regression of currency risk on trade. Appendix Tables A.5 and A.6 show the OLS estimates yield insignificant outcomes across all four risks.

Overall, both analyses provide direct evidence of how exchange rate risk falls when the goods markets of two countries become more integrated and thus closer. The general takeaway is that when trade increases by 50 percent, the risk falls by about a third of its standard deviation across countries. The effects are stable across specifications,

persist up to ten years, and are economically significant.

1.5. Isolating systematic risk

In Section 1.1.1, we showed how the risk in bilateral exchange rates depends on the relative exposure between two countries, $\varphi_i - \varphi_j$, to the global shock source of aggregate risk, u_g . The four measures of risk introduced in Section 1.1.2 capture the differential exposure to the aggregate shock based on observable quantities. Nevertheless, they are affected by the country-level or idiosyncratic component of exchange rate risk, δ_i .

To better purge this idiosyncratic risk, we isolate the systematic component of bilateral exchange rates using a principal component analysis. Formally, we extract the principal components from the vector of currency base factors:⁸

$$\Delta base_{j,t} = \underbrace{\sum_{\ell=1}^6 \beta_{j,t}^{(\ell)} PC_{\ell,t}}_{\text{Systematic Component}} + \underbrace{\sum_{m>6} \beta_{j,t}^{(m)} PC_{m,t}}_{\text{Country-level Component}} \quad (14)$$

We choose the first six principal components of the base factor returns as the systematic component of bilateral exchange rates. These first six components account for two-thirds of the variation in exchange rates. Our results are robust to including fewer components.

⁸ We decompose base factors across countries for each year because of the unbalanced nature of our panel. The weights of the decomposition can therefore vary over time.

Table 5
Second Stage: Effect of Trade on Exchange Rate Risk

	Panel A: Volatility, σ					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \log \text{Import}$	-0.018*** (0.007)	-0.037*** (0.011)	-0.024*** (0.007)	-0.024*** (0.007)	-0.018*** (0.006)	-0.012** (0.005)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.004** (0.002)	-0.010*** (0.003)	-0.007*** (0.002)	0.002*** (0.001)	0.000 (0.000)	0.000 (0.000)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,777	33,777	33,020	20,223	20,223	19,581
First stage F-statistic	15.9	10.8	13.5	10.3	10.4	10.7

	Panel B: Unshared Risk, ρ					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \log \text{Import}$	-0.273*** (0.088)	-0.434*** (0.129)	-0.277*** (0.084)	-0.375*** (0.106)	-0.303*** (0.092)	-0.209*** (0.072)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.064*** (0.022)	-0.118*** (0.038)	-0.076*** (0.025)	0.037*** (0.012)	-0.001 (0.006)	-0.001 (0.005)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,777	33,777	33,020	20,223	20,223	19,581
First stage F-statistic	15.9	10.8	13.5	10.3	10.4	10.7

This table shows the instrumental variable regressions of volatility (Panel A) and unshared risk (Panel B) on the change in imports, instrumented using trade deals. Columns (1) through (3) report the five-year change in volatility and columns (4) through (6) the 10-year change. Controls include pre-trends, country-year fixed effects for both countries, gravity variables in columns (1) and (4), and country-pair fixed effects in columns (2), (3), (5), and (6). In columns (3) and (6) we remove pegged currencies. We report standard errors clustered at the country-pair level in parentheses and the first-stage Cragg-Donald F-statistics. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

From the decomposition of base returns into systematic and idiosyncratic components, we first reformulate the four measures of exchange rate risk in Appendix A.2 and present summary statistics in Appendix Table A.2. In general, there is more variation in the systematic measures than in the idiosyncratic ones.

We present the results for the systematic risk measures in Table 7 for the staggered difference-in-difference regression and in Table 8 for the two-stage least squares regressions. We find that the systematic component responds strongly to a trade deal and, reassuringly, the effects are comparable in magnitude to those found in Table 4. For one example among many, volatility drops by 1.2 percentage points after 10 years, which is contained in the range of estimates recorded before.

In Appendix Tables A.7 and A.8, we produce a similar table but for the idiosyncratic component. Tellingly, the idiosyncratic component of exchange rate risk does not respond to trade agreements and we find that the changes in idiosyncratic measures of risk are generally neither quantitatively nor statistically significant.

Overall, our analysis here underscores that the decline in exchange rate risk is driven by the systematic component and not the idiosyncratic one. Loadings on the global factor fall following a trade deal rather than a general reduction in idiosyncratic or country-level risk.

1.6. Core and peripheral countries

Section 1.4 presented evidence concerning average effects across arbitrary countries agreeing on a FTA or a CUCMECU. We extend our analysis to see if deals have a different impact on currency risk depending on the countries' location in the global trade network. We calculate a trade centrality index for each country i at date t following Richmond (2019):

$$v_{it} = \sum_{j=1}^N \frac{\tilde{X}_{ijt} + \tilde{X}_{jit}}{\tilde{G}_{it} + \tilde{G}_{jt}} \tilde{s}_{jt}, \quad (15)$$

where $\tilde{X}_{ij}$ is the total exports from country i to j ; $\tilde{G}_j$ is j 's GDP; and $\tilde{s}_j$ is j 's global share of exports in percent— j 's total exports divided by all countries' total exports. Countries have a large index if they have a high trade intensity, $(\tilde{X}_{ijt} + \tilde{X}_{jit})/(\tilde{G}_{it} + \tilde{G}_{jt})$, with partners responsible for a large fraction of global trade.

Based on this index and depending on their relation to the median country's, we classify countries as peripheral (below) and core (equal or above) every year in our sample. We then run our two-stage least squares regression for each of the two groups and present our results in Appendix Tables A.9 to A.12.

For peripheral countries, the results show statistically significant reductions in risk for every specification. The

Table 6
Second Stage: Effect of Trade on Common Factor Regressions

	Panel A. Base Loading, β					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.264** (0.113)	-0.393*** (0.141)	-0.277** (0.112)	-0.436*** (0.130)	-0.358*** (0.124)	-0.278** (0.112)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.060** (0.028)	-0.105** (0.041)	-0.073** (0.032)	0.043*** (0.015)	-0.000 (0.008)	0.000 (0.006)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,777	33,777	33,020	20,223	20,223	19,581
First stage F-statistic	15.9	10.8	13.5	10.3	10.4	10.7

	Panel B. Share of Systematic Risk, R^2					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.134** (0.053)	-0.228*** (0.076)	-0.147*** (0.055)	-0.200*** (0.061)	-0.175*** (0.058)	-0.124** (0.049)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.033** (0.013)	-0.063*** (0.022)	-0.041*** (0.016)	0.020*** (0.007)	-0.000 (0.004)	0.000 (0.003)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,745	33,745	32,990	20,197	20,197	19,557
First stage F-statistic	15.9	10.7	13.5	10.3	10.4	10.7

This table shows the instrumental variable regressions of the slope coefficient (Panel A) and R-squared (Panel B) from [equation \(4\)](#) on the change in imports, instrumented using trade deals. Columns (1) through (3) report the five-year change in volatility and columns (4) through (6) the 10-year change. Controls include pre-trends, country-year fixed effects for both countries, gravity variables in columns (1) and (4), and country-pair fixed effects in columns (2), (3), (5), and (6). In columns (3) and (6) we remove pegged currencies. We report standard errors clustered at the country-pair level in parentheses and the first-stage Cragg-Donald F-statistics. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

magnitudes are particularly large in the first five years. For example, following a 50 percent increase in trade five years after a deal, the decline in R -squared reaches up to 0.145, closer to a half of a standard deviation. The coefficients for the ten-year horizon are lower, but they retain their statistical significance. For core countries, in contrast, the effects become more statistically evidence at the ten-year horizon. We conclude that trade deals have a more impactful and hastened effect in reducing the systematic exchange risk for peripheral countries in the global trade network.

1.7. Impact of deal type and robustness

Our empirical strategy relies on the narrow impact of trade agreements and trade. Our exclusion restriction assumes that trade deals solely affect exchange rates through their impact on trade. International trade agreements that stipulate other provisions, such as the alignment of monetary policy, could invalidate this restriction; for example, it is the case for admission into the eurozone. Even though we omit eurozone countries in our main gravity regression analysis, for robustness we analyze the impact that the type of deal has on trade and the risk of bilateral exchange rates.

First, we focus on bilateral agreements. Across the 221 bilateral trade deals in our sample we find that only two coincide with a change in monetary policy as described by the currency anchor classification of [Ilzetzi et al. \(2019\)](#).⁹ In [Appendix Tables A.13 through A.16](#), we confirm that the magnitudes of point estimates remain similar to our benchmark when we restrict our analysis to only bilateral deals, including whether they are accompanied by a change in anchor, situations where the confounding effects of monetary policy are likely relevant. Statistical significance largely remains across volatility, unshared risk, and R -squareds despite a fewer number of observations, although the impact on base loadings becomes insignificant at the five-year horizon. Thus, because our restricted analysis does not change the conclusions drawn from our benchmark results, we think it unlikely that the decline in the risk of bilateral exchange rates is driven by other arrangements bundled in a trade deal.

We further split deals into PTAs (both OWPTAs and TWPTAs), FTAs, and CUCMECUs and examine how the three

⁹ In 1983 Australia and New Zealand signed a trade deal and Australia stopped pegging to the U.S. dollar and New Zealand changed its anchor from the US dollar to Australian dollar. In 1986 the US and Israel signed a deal and Israel started to anchor its currency to the dollar.

Table 7

Time Effects of Trade Deal Change on the Systematic Risk Measures. Staggered Difference-in-Difference Regression.

Timing	Log Import		Volatility, σ		Unshared Risk, ρ		R-squared		Base Loadings, β	
	β	s.e.	β	s.e.	β	s.e.	β	s.e.	β	s.e.
-10	-0.072	(0.052)	0.002	(0.002)	0.072***	(0.024)	0.050***	(0.018)	0.225**	(0.091)
-9	-0.069	(0.050)	-0.000	(0.002)	0.050**	(0.023)	0.037**	(0.017)	0.180**	(0.080)
-8	-0.099**	(0.049)	-0.001	(0.002)	0.021	(0.023)	0.021	(0.017)	0.135**	(0.067)
-7	-0.126***	(0.043)	0.002	(0.002)	0.026	(0.021)	0.025	(0.016)	0.147**	(0.059)
-6	-0.086**	(0.042)	0.001	(0.002)	0.044**	(0.019)	0.035**	(0.014)	0.138***	(0.050)
-5	-0.064*	(0.038)	-0.003	(0.004)	0.032*	(0.018)	0.022	(0.014)	0.066	(0.046)
-4	-0.068**	(0.034)	0.001	(0.003)	0.048***	(0.015)	0.028**	(0.012)	0.077*	(0.040)
-3	-0.069**	(0.029)	0.001	(0.003)	0.043***	(0.014)	0.027**	(0.012)	0.090**	(0.036)
-2	-0.023	(0.025)	0.004*	(0.002)	0.040***	(0.014)	0.025**	(0.011)	0.077***	(0.027)
-1	0.011	(0.018)	0.004***	(0.001)	0.040***	(0.008)	0.029***	(0.006)	0.076***	(0.017)
1	0.039**	(0.019)	-0.000	(0.001)	-0.018***	(0.006)	-0.007	(0.005)	-0.026***	(0.009)
2	0.067***	(0.025)	0.000	(0.001)	-0.007	(0.008)	-0.001	(0.006)	-0.022	(0.015)
3	0.085**	(0.034)	-0.001	(0.001)	-0.010	(0.010)	-0.001	(0.008)	-0.030	(0.021)
4	0.104***	(0.035)	-0.002*	(0.001)	-0.015	(0.013)	-0.004	(0.009)	-0.045*	(0.023)
5	0.114***	(0.037)	-0.003*	(0.002)	-0.012	(0.014)	-0.002	(0.010)	-0.048*	(0.026)
6	0.124***	(0.044)	-0.003*	(0.002)	-0.027	(0.018)	-0.008	(0.012)	-0.063**	(0.030)
7	0.207***	(0.051)	-0.005***	(0.002)	-0.058***	(0.020)	-0.024*	(0.013)	-0.106***	(0.033)
8	0.216***	(0.055)	-0.014***	(0.002)	-0.190***	(0.026)	-0.091***	(0.017)	-0.273***	(0.038)
9	0.200***	(0.057)	-0.013***	(0.002)	-0.143***	(0.024)	-0.068***	(0.016)	-0.270***	(0.057)
10	0.243***	(0.061)	-0.012***	(0.002)	-0.125***	(0.025)	-0.055***	(0.016)	-0.247***	(0.058)
Observations	44924		48471		48353		48457		48471	
R-squared	0.841		0.938		0.285		0.528		0.667	

This table reports coefficients from a staggered difference-in-difference regression as in (12). The risk measures have been purged of idiosyncratic risk via a principal components analysis (see Section 1.5). The deal dummy equals one if countries i and j share either an FTA or a CUCMECU and zero otherwise. The set of control variables X_{ijt} includes gravity variables, time fixed effects, and the deal dummy. All time coefficients are deviations in y_{ijt} from the exact year of the trade deal change.

types of deals affect the risk of exchange rates. We depict the results from our staggered difference-in-difference specification across the three types of deals in Fig. 3. We find that, in line with the first stage, only FTAs and CUCMECUs affect the drop of exchange rate risk. The magnitude of the reduction in risk across FTAs and CUCMECUs is comparable and statistically significant. On the other hand, PTAs show no discernible effect on either trade or the risk of exchange rates.

In Appendix Figs. A.1 and A.2, we assess whether further split these types of deals into whether they are multilateral or bilateral. Multilateral deals display the same responses as our benchmark with a large and significant drop in risk after a CUCMECU or FTA with no effect detected following a PTA. Within bilateral deals, we find no effect for PTAs or CUCMECUs and while the point estimates of the change in risk after an FTA remain negative, they are only marginally significant. Overall, we conclude that detectable reductions in exchange rate risk are confined to CUCMECUs and, especially, FTAs.

Lastly, it is important to note that our method is not immune to reverse causality, though it is unlikely to affect our estimates. Because the risk of exchange rates will impact trade through second moments, it is unlikely to be large.¹⁰

¹⁰ Chaney (2016) shows how the level of exchange rate can have muted effect on trade. The complete analysis of the second moment of exchange rates on trade is beyond the scope of this paper.

2. A gravity model for international finance

We now turn to a theoretical model of international trade and exchange rates. Our model captures the essence of our empirical findings and allows for a quantitative investigation of the role trade frictions have on the structure of exchange rate risk. First, we present a standard framework in the international trade literature that ties trade to relative prices (Anderson and van Wincoop (2003) for example). In Section 3 we structurally estimate trade costs and analyze the model and in Section 4 we explore the model's predictions in two counterfactual experiments.

For clarity, we consider an Armington model where each country produces one differentiated good and consumer demand originates from constant elasticity of substitution (CES) preferences (see Armington (1969) and Anderson (1979)). While the Armington assumption abstracts from natural specialization due to comparative advantage, it gives a good description of aggregate bilateral imports and most importantly provides a transparent connection to exchange rates. The model's equilibrium delivers a tight link between trade barriers and trade flows in the form of a gravity equation. Countries that are far from the rest of the world face large resistances to trade and are isolated from it.

To understand the behavior of exchange rates we introduce productivity shocks at the country-level which, coupled with households' risk aversion, introduces risk into the international economy. We also depart from complete international asset markets and disallow nations from fully sharing the risk they face. Collectively, these essential el-

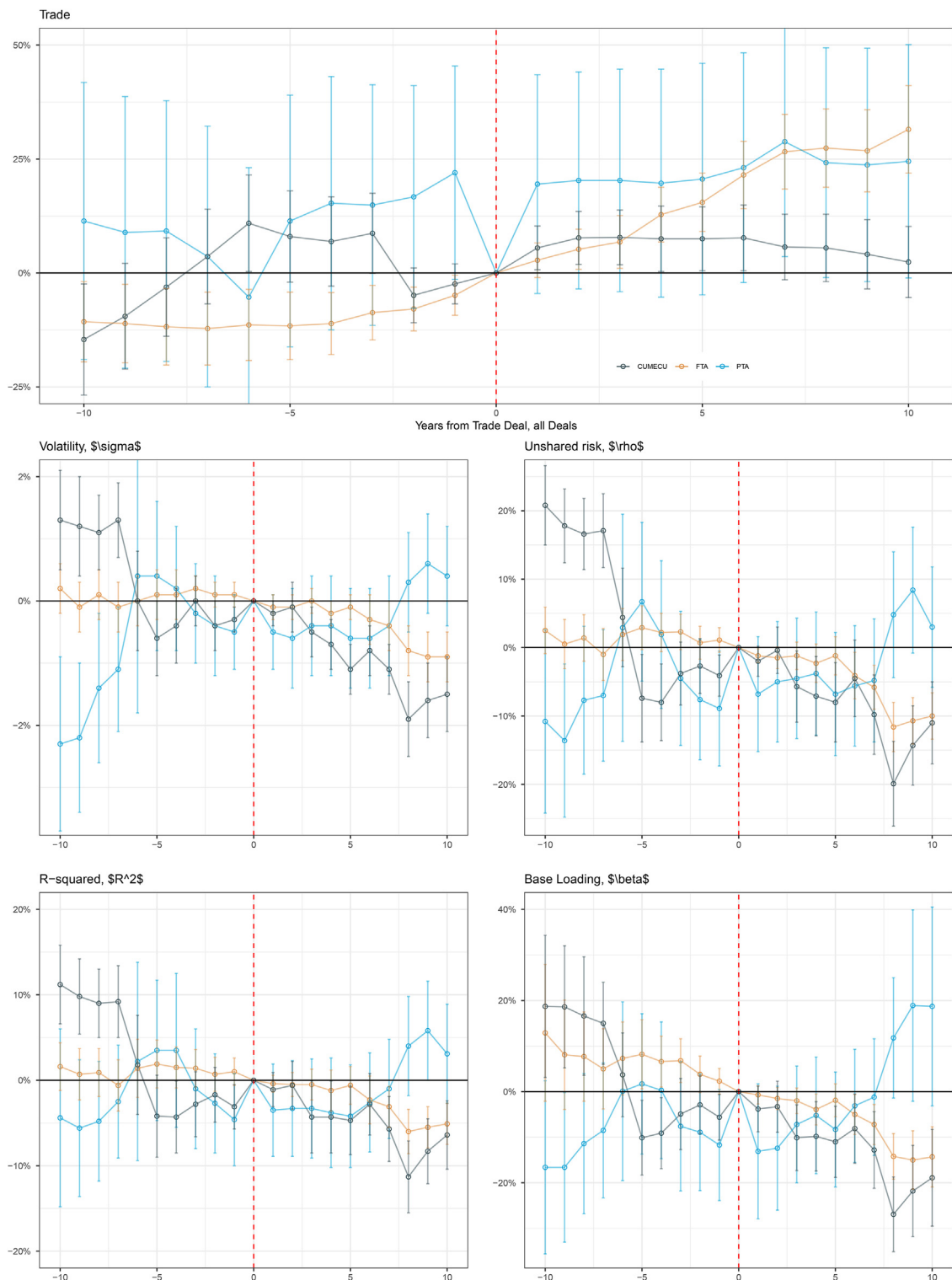


Fig. 3. Change in Log Imports and Risk Around Trade Agreements By Deal Type. This figure plots the estimates of our staggered difference-in-difference regression in (12) with trade as the dependent variable. We compare for each year around a trade deal country-pairs with and without an agreement. The regression includes domestic and foreign country-year fixed effects and the standard gravity controls described in Table 2. Our trade deal variable equals one if countries have a FTA (yellow line), a CUMECU (gray line) or a PTA (blue line) and is zero otherwise.

Table 8

Second Stage: Effect of Trade on Common Factor Regressions for the Systematic Risk Measures

	Panel A					
	Volatility, σ			Unshared Risk, ρ		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \log \widehat{\text{Import}}$	-0.017** (0.009)	-0.021** (0.011)	-0.021** (0.011)	-0.220* (0.126)	-0.180 (0.135)	-0.196 (0.135)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.002 (0.001)	-0.004* (0.003)	-0.004* (0.003)	-0.030 (0.020)	-0.036 (0.031)	-0.040 (0.031)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	18,312	18,312	17,959	18,264	18,264	17,911
First stage F-statistic	5.97	5.11	5.28	5.94	5.07	5.24

	Panel B					
	Base Loading, β			Share of Systematic Risk, R^2		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \log \widehat{\text{Import}}$	-0.385** (0.196)	-0.220 (0.183)	-0.231 (0.181)	-0.164* (0.085)	-0.162 (0.100)	-0.167* (0.100)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.056* (0.031)	-0.045 (0.041)	-0.046 (0.041)	-0.020 (0.015)	-0.031 (0.022)	-0.032 (0.022)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	18,312	18,312	17,959	18,312	18,312	17,959
First stage F-statistic	5.97	5.11	5.28	5.97	5.11	5.28

This table shows the instrumental variable regressions of five-year changes in volatility and unshared risk (Panel A) and base loadings and R -squared (Panel B) from equation (4) on the change in imports, instrumented using trade deals. The risk measures have been purged of idiosyncratic risk via a principal components analysis (see Section 1.5). Controls include pre-trends, country-year fixed effects for both countries, gravity variables in columns (1) and (4), and country-pair fixed effects in columns (2), (3), (5), and (6). In columns (3) and (6) we remove pegged currencies. We report standard errors clustered at the country-pair level in parentheses and the first-stage Cragg-Donald F-statistics. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

ements tie trade flows and exchange rates to provide a theoretical foundation of how trade deals affect the factor structure of exchange rates.

2.1. Preferences and technology

There are N countries indexed by i, j, k . To fix ideas, country j is the base country and imports goods from country i and exports to k . Country j has population L_j and an infinitely-lived representative household with time-separable utility with constant relative risk aversion $\gamma > 0$ and rate of time preference $\rho > 0$,

$$U_j = \mathbf{E} \left[\int_0^\infty e^{-\rho t} u(C_j(t)) dt \right], \quad (16)$$

where $u(C) = C^{1-\gamma}/(1-\gamma)$ and labor is employed linearly to produce the country's tradable good with productivity $z_j(t)$.

Countries can trade bilaterally subject to iceberg transportation costs: in order for one unit of i 's good to be imported by j , $\tau_{ij} \geq 1$ units must be shipped. We place two restrictions on the trade cost matrix: all diagonal elements equal one, $\tau_{jj} = 1$ for all j ; and the triangle inequality, $\tau_{ik} \leq \tau_{ij} \tau_{jk}$ for all i, j, k , which implies that it is never

cheaper to ship a good via an intermediate location rather than sell directly to a destination.¹¹

Consumers have homothetic preferences over the set of goods produced in countries across the world. Households' consumption C_j is a CES aggregator of all goods imported, $q_{ij \neq j}$, and produced locally, q_{jj} ,

$$C_j(t) = \left(\sum_{i=1}^N (q_{ij}(t))^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}. \quad (17)$$

The parameter $\eta > 1$ is the elasticity of substitution between individual goods.

2.2. Asset markets and equilibria

In addition to goods market frictions, the degree of international risk sharing co-determines equilibrium alloca-

¹¹ The trade cost specification is equivalent to the shipping industry being competitive, working at zero profits and solely covering the cost of resources. Greenwood and Hanson (2015) and Ready et al. (2017b) show shipping dynamics influence trade costs and carry trade profits. While the presence of hub-and-spoke networks in global trade suggest that the triangle inequality is not an innocuous assumption and that temporal lags in shipping could be important, we maintain it to impose the no-arbitrage condition present in virtually all of the international finance literature.

tions. We examine equilibria under polar opposite international asset market structures: financial autarky and complete markets. Under financial autarky we find the competitive equilibrium, and with complete markets we solve the planner's problem.

We describe our approach to modeling intermediate cases of imperfect risk sharing in Section 3.1.1. While international markets may not be complete, our representative agent assumption requires us to have complete markets within a country.

Our economy is static and can be cast as a sequence of one-period problems. For notational clarity we ignore dependence on time t when there is no ambiguity or importance.

2.2.1. Financial autarky

In financial autarky, households are precluded from trading in international asset markets to smooth consumption. We discuss the firms' problems before turning to households.

Firms maximize profits and choose the optimal quantity of labor to hire given the wage rate w_j . Markets are competitive and firms set the domestic price of the tradable good to $p_{jj} = w_j/z_j$.

Let p_{ij} denote the exporter's free on board (f.o.b.) price in country i . Then on a cost-insurance-freight (c.i.f.) basis, we have $p_{ij} = p_{ii}\tau_{ij}$. For each good shipped, from i to j the exporter incurs exports costs equal to $\tau_{ij} - 1$ of country i goods. The exporter passes these costs on to the importer. Thus, the value of imports of j from i is $p_{ij}q_{ij}$, the origin's production value, $p_{ii}q_{ij}$, plus the trade cost $(\tau_{ij} - 1)p_{ii}q_{ij}$ that the exporter passes on.

At each point in time, country j takes prices $\{p_{ij}\}_i$ as given and chooses consumption C_j and the demand for goods $\{q_{ij}\}_i$ to maximize (16) subject to (17) and its budget constraint

$$\sum_{i=1}^N p_{ij}q_{ij} = w_jL_j. \quad (18)$$

We close the model by solving for $\{w_j\}_j$ with the global resource constraint on tradable goods

$$z_jL_j = \sum_{k=1}^N \tau_{jk}q_{jk}, \quad \text{for all } j. \quad (19)$$

Walras' law implies that the equilibrium is unique only up to normalization. We normalize w_j for the United States to equal 1.

Appendix B gives the complete derivation of the first-order conditions. Under financial autarky, consumption is tightly linked to output, $p_jC_j = w_jL_j$, and trade is balanced as the value of total imports including domestic absorption, $\sum_i p_{ij}q_{ij}$, equals the value of total exports $p_{jj}z_jL_j$ for every country.

2.2.2. Complete markets

With a complete set of tradable state contingent claims, the central planner chooses an optimal allocation of consumption $\{C_j\}$ and goods $\{q_{ij}\}_i$ for a suitable choice of Pareto weights $\{\lambda_j\}$ for each country's representative

household

$$\mathbf{E} \left[\sum_{j=1}^N \lambda_j \int_0^\infty e^{-\rho t} u(C_j(t)) dt \right], \quad (20)$$

subject to every country's resource constraint (17) and the global resource constraint (19). We follow Fitzgerald (2012) and interpret λ_j as the inverse of the marginal utility of wealth for country j which equals the inverse of the multiplier on the country's budget constraint in the decentralized economy.

In contrast to financial autarky, complete markets allow trade to be unbalanced and to let countries smooth their consumption over time by deviating it from their output ($p_jC_j \neq w_jL_j$, in general). In particular, a marginal increase of q_{ij} trades off an improvement of j 's marginal utility with a tightening of i 's global resource constraint.

2.3. Exchange rates and gravity

In equilibrium, household's j demand for differentiated goods follows the CES demand system,

$$q_{ij} = p_{ij}^{-1} \cdot \left(\frac{p_{ij}}{P_j} \right)^{1-\eta} \cdot P_j C_j, \quad (21)$$

which clarifies the determinants of the demand for good q_{ij} : a direct price effect, p_{ij} ; a substitution effect through the local price index, $(p_{ij}/P_j)^{1-\eta}$; and an income effect, $P_j C_j$.

In contrast to a Cobb-Douglas specification where $\eta = 1$, if the price of good i , p_{ij} , increases relative to the price index in country j , P_j , households will substitute away from q_{ij} to all the other goods in their consumption basket $\{q_{kj}\}_{k \neq i}$. Thus, a change in a single bilateral trade cost will affect, through substitution, all countries' consumption allocations.

This is evident from the expression for the price index in country j ,

$$P_j = \left[\sum_{i=1}^N (\tau_{ij} p_{ii})^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad (22)$$

which depends on the trade costs of all imported goods and doubles as an index of inward bilateral trade costs. Greater trade barriers, and thus higher average prices, reduce a country's consumption for a given income.

The definition of the real exchange rate S_{ij} of i in units of j is standard and equals the ratio of consumption price levels

$$S_{ij}(t) = \frac{\lambda_j(t)u'(C_j(t))}{\lambda_i(t)u'(C_i(t))} = \frac{P_j(t)}{P_i(t)}. \quad (23)$$

Under this ratio, if S_{ij} rises, then j 's currency appreciates relative to i 's. We emphasize here with the dependence on time t that relative marginal utilities of consumption fluctuate for two reasons. First, asset market frictions imply that the relative marginal utility of wealth, $\lambda_j(t)/\lambda_i(t)$, can vary over time along with output. Second, because trade costs impede risk sharing regardless of asset market structure, relative utility will not be equal. Instead, the real exchange rate, $P_j(t)/P_i(t)$, a summary of relative trade costs,

will comove with output. If for example τ_{ij} were equal to one for all i, j , then P_j would become identical across all countries and, in the absence of financial market imperfections, the global economy would achieve perfect risk sharing.

2.3.1. Gravity and multilateral resistance

The gravity framework places inward and outward multilateral resistance as central in determining trade shares and exposures to shocks to exchange rates. We defined inward resistance in (22) and here we define outward resistance by first multiplying the global resource constraint in (19) by p_{jj} to give

$$w_j L_j = p_{jj}^{1-\eta} \Pi_j^{1-\eta}. \quad (24)$$

Country j 's index of outward bilateral trade costs, $\Pi_j^{1-\eta} = \sum_{k=1}^N p_k^{\eta-1} \tau_{jk}^{1-\eta} w_k L_k$, can be thought of as if the country exported its product to a single world market facing a global trade cost of Π_j .

Moving on, by multiplying the CES demand equation by p_{ij} and substituting in (24) we obtain the gravity equation

$$p_{ij} q_{ij} = w_i L_i \times P_j C_j \times \left(\frac{P_j \Pi_i}{\tau_{ij}} \right)^{\eta-1}. \quad (25)$$

The value of imports, $p_{ij} q_{ij}$, depends on the product of countries' outputs ($w_i L_i \times P_j C_j$) and terms due to trade costs. A higher bilateral trade barrier, τ_{ij} , has a direct effect of reducing trade between the two countries. An increase in outward resistance faced by the exporter, Π_i , is accompanied by a decline in its supply price, p_{ii} ; for a given τ_{ij} , this grows trade between the two countries. Finally, a rise in the importer's inward resistance lowers the relative price of goods from i , leading to greater trade. Country j 's imports from i depend on all resistances, $\{\tau_{ij}\}$, that, in turn, determine the cross-section of currency exposure.

2.4. Risk and currency exposure

We focus on a stationary economy and assume that the vector of productivity for the N countries in the global economy follows a mean-reverting process in log-arithmetic. We collect every productivity into a N -vector $z = (z_1, \dots, z_N)'$ and write the stochastic process

$$d \log z = -\kappa \log z dt + \nu dB, \quad (26)$$

where $\kappa > 0$ is the rate of mean reversion, ν is a scalar that captures the volatility of productivity, and B is a vector of Brownian motions in $\mathbb{R}^N$ that are independent of each other. With the independence of productivity shocks, trade across countries is their only link and the correlation of countries' business cycles will be fully determined by the structure of trade barriers.

The log change of currency S_{ij} follows immediately, $d \log S_{ij} = d \log P_j - d \log P_i$, and as before the definition of the base factor for country j is the average appreciation of its currency vis-à-vis all others, $d \text{base}_j = d \log P_j - \frac{1}{N} \sum_{i=1}^N d \log P_i$. It is then straightforward to calculate in our

model simulations the analogs of our empirical measures of exchange rate risk:

$$\begin{aligned} \sigma_{ij}^2 &= \text{Var}(d \log P_j - d \log P_i) \\ \beta_{ij} &= \frac{\text{Cov}(d \log P_j - d \log P_i, d \text{base}_j)}{\text{Var}(d \text{base}_j)} \\ R_{ij}^2 &= \beta_{ij}^2 \times \frac{\text{Var}(d \text{base}_j)}{\text{Var}(d \log P_j - d \log P_i)} \\ \rho_{ij} &= \frac{\text{Var}(d \log P_j - d \log P_i)}{\text{Var}(d \text{base}_j) + \text{Var}(d \text{base}_i)}. \end{aligned}$$

These measures are exact and do not rely on the large N limit required by their empirical counterparts. Consequently, our gravity model naturally generates aggregate shocks from origins that are granular (Gabaix, 2011).

2.4.1. Volatility approximation and exposure to global shocks

We can use our theory to look into the drivers of global shock exposures, something not directly observed in the data. Our general equilibrium analysis, however, prevents closed-form solutions to goods' prices. Nevertheless, we can progress by exploiting our continuous-time setup that allows us to write these prices as following diffusion processes

$$dp_{jj} = \mu_j(\{p_{jj}\}_j) dt - \sigma_j(\{p_{jj}\}_j) dB, \text{ for all } j. \quad (27)$$

In writing this, we highlight the dependence of p_{jj} 's drift and volatility on all prices and shocks (and conditional on parameters) and note a negative productivity shock raises prices.

For each simulation path of our stationary economy, we calculate numerically the unconditional volatility of price changes for each country's good as the left side of (27). We label this estimate $\hat{\sigma}_j$ where we are implicitly defining a system of price processes

$$dp_{jj} = \mu_j(\{p_{jj}\}_j) dt - \hat{\sigma}_j d\hat{B}_j, \text{ for all } j, \quad (28)$$

that are driven by the collection of Brownian motions $(\hat{B}_1, \dots, \hat{B}_N)$ that need not be independent.

The risk in a bilateral exchange rate is determined by the dynamics of i and j 's consumption prices in (22) that, in turn, are formed by their underlying exposures to shocks. To better understand these structural drivers of risk, we use Ito's lemma and our approximation in (28) to derive expressions of volatility

$$\begin{aligned} d \log P_j &= \mu^P_j dt - \sum_{i=1}^N \tau_{ij}^{1-\eta} P_j^{\eta-1} p_{ii}^{-\eta} \hat{\sigma}_i d\hat{B}_i \\ &= \mu^P_j dt - \sum_{i=1}^N \omega_{ij} \frac{\hat{\sigma}_i}{p_{ii}} d\hat{B}_i, \end{aligned} \quad (29)$$

where the second equality multiplied and divided by p_{ii} and substituted in (24). The drift term, μ^P_j , is unimportant for what follows and consequently not written out. The volatility term, however, expresses country j 's exchange rate exposure to i as a weighted function of price volatility and prices:

$$\varphi_{ij} = \omega_{ij} \cdot \frac{\hat{\sigma}_i}{p_{ii}} = \frac{p_{ij} q_{ij}}{P_j C_j} \cdot \frac{\hat{\sigma}_i}{p_{ii}}. \quad (30)$$

Importantly, the weights ω_{ij} , which sum to one across i , correspond to import shares. We emphasize that the volatility approximation $\hat{\sigma}_i$ is only used when calculating φ_{ij} in our simulations.

Equations (29) and (30) are intuitive—exposure in import shares translates into exposure in exchange rates—and provide a powerful connection between international trade and finance. Country j 's exposure, φ_{ij} , closely relates to the multilateral resistance terms defined in (25) and therefore to the standard gravity equation. As in gravity, larger direct bilateral trade costs attenuate exposure while higher levels of inward or outward resistance amplify it for similar reasons. Exposure, moreover, is greater to larger and highly productive countries, $w_i L_i / p_{ii} = z_i L_i$, as well as those with more price variability, $\hat{\sigma}_i$.

Equation (30), furthermore, provides an endogenous definition of a core country if $\bar{\varphi}_j = \sum_{i \neq j} \varphi_{ij} / (N - 1) > \bar{\varphi}$ and peripheral if $\bar{\varphi}_j < \bar{\varphi}$ as $\bar{\varphi} = \text{median}(\bar{\varphi}_j)$ is determined within equilibrium. Taking a step back, this expression reproduces the core-periphery structure common to many modern models of international finance. Core countries disproportionately bear more global risk. In bad times, core countries' marginal utilities rise relatively more and their currencies appreciate. From their perspective, peripheral countries' currencies appear risky *ex ante* as they depreciate in bad times. For example, Hassan (2013) shows large countries are core; Richmond (2019) identifies core countries that serve as hubs to the global trade network, like Netherlands and Singapore.¹² We provide a new link based on gravity, thus tying our work closely to the literature in international trade.

3. Structural estimation and main model predictions

Having described the theoretical model in Section 2, we now turn to structurally estimate trade costs and evaluating model predictions before using it to examine counterfactuals in Section 4.

3.1. Structural estimation of trade costs

We conduct our structural estimation under financial autarky, as this asset structure obviates the choices of a planner's Pareto weights and risk aversion. We do, however, need to choose productivity levels and the elasticity across goods, η . We put productivity at $z_j = 1$ for all j . Extensive prior work in the realm of international trade places the elasticity parameter in the range of five to ten (Anderson and van Wincoop (2004)) and we set $\eta = 6$.¹³

¹² Corte et al. (2016) show net creditors of external liabilities as core, Ready et al. (2017a) depict core countries as those that tend to export final goods and import commodities, and Colacito et al. (2018b) link core exposure to global growth news shocks.

¹³ Armington and CES assumptions are widely used in the trade literature in spite of a well-known shortcoming: the elasticity of substitution across goods needs to be high to have trade costs materially affect trade. A more recent strand of work (for example Melitz (2003) and Chaney (2008), among others) identifies the size or productivity distribution of firms as material to trade barriers and trade and creates an important role for the extensive margin to affect trade. All of these models, however, generate similar implications for welfare Arkolakis et al. (2012).

Rather than employ the gravity equation, which is currency dependent, we use import shares, ω_{ij} defined in (30), in our estimation that is normalized by the output of the importing country. Normalizing by output makes shares invariant to sample selection, whereas normalizing by the sample's total imports would not. We normalize the US's GDP to one by setting L_j to one. (Recall that $w_j = 1$ for $j = \text{USA}$; hence $w_j L_j = 1$.)

Given this setup, we estimate τ_{ij} for all i and j by minimizing the distance between the model's and the data's import shares and GDP with an algorithm based on tatonnement. During estimation, we enforce the diagonal of the matrix to one and the triangle inequality: for every i, j, k , we find the minimum triad across τ_{ij} , τ_{ik} , and τ_{jk} and set the minimum to satisfy the inequality by equating it with the sum of triad's other two elements.

Fig. 4 depicts the outcome. Indirectly targeted moments are domestic absorption, $\omega_{jj} = 1 - \sum_{i \neq j} \omega_{ij}$, and imports in US dollars. All targets, both direct and indirect, possess high correlations of 0.87 and over as seen in each panel. A discrepancy between model and data is that in the data imports are gross measures while GDP is value-added, so in the data domestic absorption can be negative.¹⁴

We summarize the trade cost estimates in Table 9. The mean of average costs, $(\sum_i \tau_{ij} - 1)/N$, is 1.75, close to the 170 percent barrier estimated in Anderson and van Wincoop (2004) as representative of all transport, border-related, and local distribution costs. In general, import and export trade costs are lower for most developed countries. Measures of resistance that adjust for equilibrium prices portray China, Japan, and the US as possessing the smallest averages of inward and outward resistance.

3.1.1. Remaining calibration and discussion of asset market structure

We now complete the remaining calibration of our model. While our economy does not grow over time, it does fluctuate with the underlying productivity shocks that result in business cycles. We set $\kappa = 0.385$ to target a half-life of output gaps of $\log(2)/\kappa = 1.8$ years, near the range provided in Chari et al. (2000) of between 2 and 2-1/2 years in the United States. We jointly set $\gamma = 7.9$ and $\sigma = 0.39$ to match consumption volatility while generating a reasonable amount of exchange rate volatility.

We then simulate our economy many times over a time period that matches the length of our data sample and average over the simulated moments. Consumption and output are time-aggregated from a monthly to an annual frequency. Table 10 summarizes our work that targets informative moments standard to asset pricing, international macroeconomics, and international finance. In constructing the four bilateral moments in Panel B, we follow Lustig and Verdelhan (2019) and create 90 percent confidence intervals (CI) based on the distribution of pair-wise statistics.

¹⁴ Previous work adjusts the import share's numerator or denominator for consistency; for example, Fitzgerald (2012) scales GDP by two for all countries to form gross output. Value-added exports represent about 73 percent of gross exports and trade in intermediate goods account for around two-thirds of international trade Johnson and Noguera (2012).

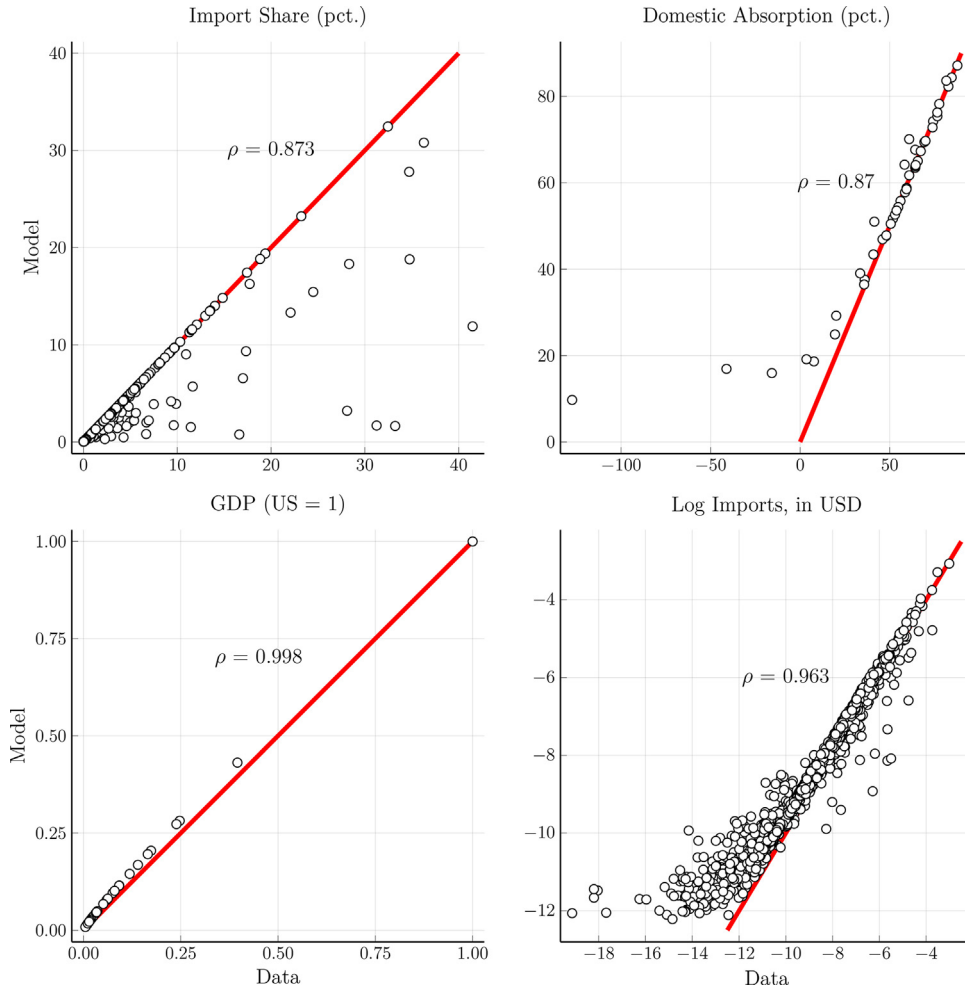


Fig. 4. Structural Estimation of Trade Costs This figure depicts the fit of the structural estimation of trade costs with a 45 degree line. The gravity model of Section 2 is solved under financial autarky and targets averages of output and import shares in the data. The correlation coefficient across model and data within each scatter is recorded by ρ .

First, complete markets generate an ordering of consumption and currency volatility that is consistent with the data. Financial autarky, conversely, fully exposes countries to productivity shocks, resulting in consumption volatility several times larger than what is observed empirically. But as stressed in Backus et al. (1992), complete markets unfortunately produce cross-country consumption correlations that exceed those of output correlations, which is counterfactual to the data. Related is the Backus-Smith puzzle: under the additional assumption of constant relative risk aversion, a regression of exchange rate changes on relative consumption growth yields a slope coefficient equal to γ . This puzzle is seen simply by rearranging (23) under constant λ_j 's:

$$d \log S_{ij} = \gamma (d \log C_i - d \log C_j) \quad (31)$$

At the heart of these puzzles is the connection between quantities and prices, which are functions of risk. Complete markets allocate risk to those countries who are best suited to bear it and, as a result, core countries often exhibit greater and more correlated variation in marginal utility than do peripheral ones. This, by itself, seems un-

intuitive as asset market frictions and volatility are problems commonly thought of as being concentrated in developing economies. Nevertheless, a well-known phenomenon of foreign exchange markets is that during bad times, the US dollar appreciates, even if those bad times originate in the US.

Several approaches in the literature have been used to resolve or at least to alleviate these puzzles. For example, Colacito and Croce (2011) employ recursive preferences, Kollmann (2012) and Hassan (2013) use restricted participation in asset markets, and Pavlova and Rigobon (2012) make markets incomplete.¹⁵ Our approach lets Pareto weights deviate linearly from the constant vector achieved under complete markets to the vector that varies under financial autarky:

$$\hat{\lambda}_j(t) = (1 - \phi_j)\lambda_j(t) + \phi_j\lambda_j, \quad (32)$$

¹⁵ Additionally, Alvarez et al. (2002), Chien et al. (2015), Dou and Verdelhan (2015) use segmented markets and Chari et al. (2000), Gabaix and Maggiori (2015), and Favilukis et al. (2015) also assume incomplete markets.

Table 9
Structurally Estimated Trade Costs by Country

	Average Import Trade Costs	Average Export Trade Costs	Inward Resistance	Outward Resistance
Australia	1.94	1.7	1.23	1.28
Austria	1.77	1.72	1.19	1.37
Belgium	0.91	1.25	1.0	1.35
Brazil	2.18	1.73	1.22	1.23
Canada	1.87	1.64	1.11	1.23
Chile	2.28	2.05	1.42	1.47
China	1.69	1.37	1.08	1.13
Colombia	2.52	2.3	1.43	1.44
Czech Republic	1.23	1.77	1.02	1.47
Denmark	1.8	1.71	1.27	1.39
Egypt	1.78	2.3	1.34	1.47
Finland	1.86	1.69	1.31	1.42
France	1.53	1.34	1.08	1.17
Germany	1.3	1.01	1.04	1.13
Greece	2.04	2.26	1.35	1.41
Hungary	1.47	1.8	1.17	1.51
India	1.99	2.05	1.23	1.25
Indonesia	2.02	2.04	1.26	1.35
Ireland	1.4	1.76	1.1	1.45
Israel	1.78	2.03	1.3	1.45
Italy	1.58	1.35	1.12	1.19
Japan	1.74	1.38	1.05	1.09
Korea	1.59	1.48	1.14	1.28
Malaysia	1.1	1.68	0.99	1.45
Mexico	2.1	2.01	1.19	1.27
Netherlands	0.9	1.32	0.97	1.29
New Zealand	2.29	2.09	1.43	1.49
Norway	1.98	1.91	1.29	1.38
Peru	2.58	2.09	1.51	1.51
Philippines	1.84	2.15	1.3	1.47
Poland	1.83	1.91	1.27	1.38
Portugal	1.88	2.08	1.3	1.44
Qatar	2.38	2.0	1.51	1.55
Russia	1.78	1.85	1.16	1.2
Singapore	1.45	1.44	1.2	1.46
South Africa	1.99	1.74	1.34	1.4
Spain	1.67	1.73	1.17	1.25
Sweden	1.7	1.47	1.22	1.34
Switzerland	1.39	1.45	1.12	1.33
Thailand	1.54	1.73	1.19	1.42
Turkey	1.94	2.08	1.28	1.34
UK	1.35	1.29	1.07	1.17
United Arab Emirates	1.48	2.0	1.22	1.43
USA	1.57	1.28	0.96	1.0
Mean	1.75	1.75	1.21	1.34
Standard Deviation	0.38	0.38	0.14	0.13

This table reports the results of the structural estimation of trade costs under financial autarky (balanced trade). The estimation of the gravity model of Section 2 targets averages of output and import shares in the data. Fig. 4 depicts the estimation's fit.

where $\phi_j \in [0, 1]$ indexes market completeness. We first calculate Planner weights as an output of the structural estimation: $\lambda_j = P_j/C_j^{-\gamma}$. We then implement (32) by solving the financial autarky model for a path of productivity shocks and then, for each realization of $\lambda_j(t)$ along the path, we resolve the complete markets economy using $\lambda_j(t)$ rather than λ_j . In this way we adopt Basak and Cuoco (1998) insight that a complicated problem of restricted stock market participation can instead be solved by allowing Pareto weights to vary over time, as in incomplete markets.

While parameterizing ϕ_j to a priori features of countries seems reasonable, our goal is simply to weaken the dependence between prices and quantities. We therefore choose a single value $\phi_j = \phi = 0.61$ that creates the correct ordering of average cross-country consumption and

output correlation and average consumption and currency volatility. With this value for ϕ , the model's average Backus-Smith coefficient also becomes contained in the data's confidence interval. In complete markets the coefficient should be identical to γ yet we see that time aggregation biases it downward.

3.2. Main model predictions

In this section, we look at the main predictions of the model that connect trade and systematic risk. They are depicted in Fig. 5.

First, we confirm that trade costs capture distance and generally increase along with the risk exposure as measured by base loadings. Panel A is a scatter plot of the average beta across simulations, β_{ij} , on the elements, τ_{ij} , of

Table 10
Summary of Calibration

Panel A: Calibration				
Description	Parameter	Value	Target/Source	
Elasticity between goods	η	6.0	Anderson and van Wincoop (2004)	
Mean reversion of productivity	κ	0.385		
Risk aversion	γ	7.9	Consumption and currency volatility	
Productivity volatility	σ	0.39		
Panel B: Moments (Annual)				
	Data (90% CI)	Financial Autarky $\phi = 0$	$\phi = 0.61$	Complete Markets $\phi = 1$
Consumption volatility	(0.007, 0.040)	0.274	0.033	0.022
Currency volatility	(0.067, 0.291)	0.035	0.117	0.119
Cross-country consumption correlation	(-0.425, 0.618)	0.007	0.355	0.887
Cross-country output correlation	(-0.162, 0.758)	0.512	0.356	0.356
Backus-Smith coefficient	(-3.715, 1.867)	0.048	1.383	5.972

This table summarizes the calibration of the gravity model outlined in Section 2. All data moments in Panel B are real and are reported as 90 percent confidence intervals obtained from the distribution of pair-wise statistics, as in Lustig and Verdelhan (2019). The model is simulated monthly over a time period that matches the length of our data sample and consumption and output are time-aggregated to annual observations. The Backus-Smith coefficient is the slope coefficient of a regression of annual changes of exchange rates on an intercept and annual changes in relative consumption growth as in (31).

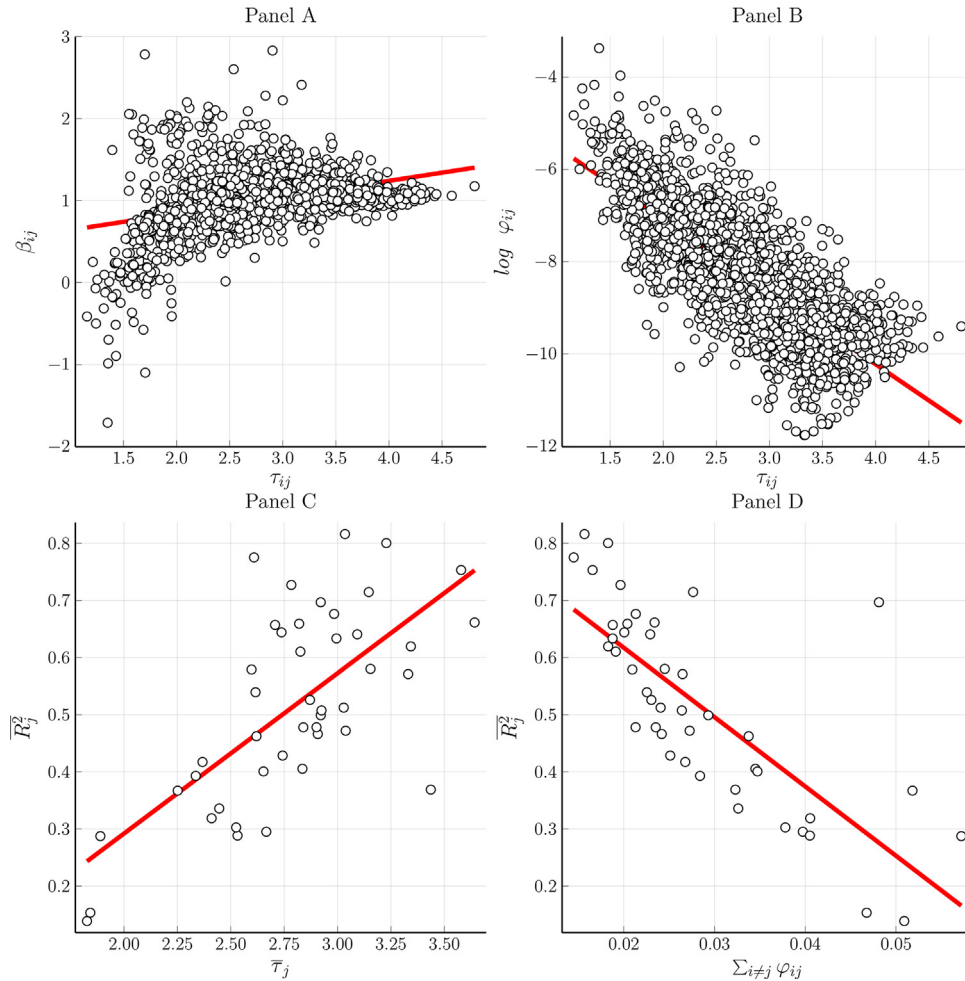


Fig. 5. Main Model Predictions of Calibrated Economy This figure shows the main predictions of the calibrated model in Section 2 under a market completeness of $\phi = 0.61$. The average inward trade cost is the average over i of trade costs of imported goods by j , $\bar{\tau}_j = \sum_{i \neq j} \tau_{ij} / (N - 1)$. The average share of outward systematic risk is from the perspective a currency investor in j , $\bar{R}_j^2 = \sum_{i \neq j} R_{ij}^2 / (N - 1)$. Total import exposure is $\sum_{i \neq j} \varphi_{ij}$.

the trade cost matrix. Countries further apart tend to have a greater difference in exposure.

In Panel B we scatter each country's import exposure (in logs), φ_{ij} , on the trade cost matrix that depicts a negative relationship. Intuitively, a country's import exposure declines with distance. The figure also shows that core countries, those with high average import exposures, generally have lower average trade costs.

As a further analytical step we calculate $\bar{R}_j^2 = \sum_{i \neq j} R_{ij}^2 / (N - 1)$. This is *outward* systematic risk, a variable that answers "How much systematic risk do I typically see looking out as a currency investor in country j ?" We see this variable positively correlates with average trade costs, $\bar{\tau}_j = \sum_{i \neq j} \tau_{ij} / (N - 1)$, in Panel C. Core countries are acutely exposed to systematic risk, so when looking out to the periphery they see little. In contrast, peripheral countries, which face high inward trade costs, see a great deal when looking into the core.

More succinctly, when core countries look outwards they see high betas but low R -squareds. When an adverse global shock hits, marginal utilities rise more in core countries, appreciating their currencies, especially with respect to peripheral economies. Thus an appreciation in j 's base factor corresponds with a big appreciation of S_{ij} for distant i —high betas. This is because peripherals are relatively less exposed to the global shock—low R -squareds.

We corroborate this narrative in Panel D where we plot outward systematic risk on j 's total import exposure, $\sum_{i \neq j} \varphi_{ij}$, a key part of exchange rate volatility in (29). Because productivity volatility ν is constant across countries, total import exposure is like a variance decomposition of exchange rates. The scatter shows that core countries tend to possess the highest total import exposures. This translates into core countries having the greatest exposure to global shocks and displaying the most correlated currency movements. Put differently, there is a large common component of exchange rate risk among core countries. Most peripheral countries' exchange rates, by contrast, are driven by country-specific shocks.

4. Counterfactuals

Having calibrated the model, in this section we run two counterfactuals. The first replicates our empirical method in the model and compares the model's and the data's predictions for systematic risk. The second counterfactual implements a US-China trade war and studies in particular how it affects peripheral countries.

4.1. Bilateral reductions in trade costs

The empirical results of Section 1 showed that a typical trade deal raises imports by 50 percent over five years and reduces systematic risk by a third of a standard deviation across countries. Here we replicate the first stage and look at the model's predictions for the second stage. Comparing in this way allows an independent test of the model's efficacy in matching the data.

We implement the first stage in the following way. For each pair of countries (i, j), we adjust their bilateral trade

costs to generate a 50 percent growth in imports to i from j and to j from i . We then simulate this "After" economy and record the changes in risk measures specific to this bilateral pair. We do this for each of the $N(N - 1)/2$ bilateral pairs.

The rows spanned by Before in Panel A in Table 11 summarizes the mean and standard deviation of exchange rate risks across countries in the model. In comparison to the empirical moments in Table 1, the model generates less dispersion than in the data, possibly because the data figures reflect changes in nominal prices not present in the real model. The ranking of variation across risk measures is consistent, however, as in both model and data factor loadings and shares of systematic risk display the greatest variation.

Rows listed after show the average change across all bilateral pairs after implementing the first stage in the model. In accordance with our empirical results, all four risk measures decline. The shifts range from 0.067 up to 0.174 of a standard deviation. In general, the shifts are less marked in the model than in the data. Our model abstracts from monetary and financial forces that perhaps reinforce the effect of trade on risk. That said, our model is best thought of measuring the impact on risk from trade directly.

In Panel B, we split the global economy into core and peripheral countries based on their relation to $\bar{\varphi}$. Bilateral pairs are either both core or both peripheral. Consistent with the empirical results in Appendix Tables A.9–A.12 the effects of a trade deal are concentrated among peripheral countries. Countries that are initially the riskiest benefit the most from the reduction in trade costs.

Why these effects concentrate on the peripheral derives from the gravity equation. The structural equation places a nonlinear relationship between multilateral resistance and import shares. Peripheral countries, which initially face the greatest trade costs, undergo the biggest drop in resistance. Altogether, the counterfactual highlights that poorer countries benefit disproportionately more than do rich countries from a trade deal. That trade has a first-order effect on currency risk should not be understated.

4.2. A US-China trade war

The recent decade has seen China becoming a budding superpower and emerging as a key player in global trade, and not without dispute. The recent trade war between the United States and China saw American average tariffs rise to 17.5 from 2.6 percent and a Chinese retaliation of going from 6.2 to 16.4 percent Bekkers and Schroeter (2020). Within categories of goods, we witnessed extremes and, as an example, watched Chinese imports of American soybeans fall to zero.

Tariffs, however, constitute only a portion of total trade costs in our model that are much more comprehensive. As a reasonable counterfactual we raise bilateral trade costs by 50 percent and see what an escalation into a severe war would look like. Table 12 tabulates the aftermath.

Panel A shows the impact on the United States and China. Import shares between countries fall drastically, but because their bilateral imports are relatively small to

Table 11
Bilateral Reduction in Trade Costs

Panel A: Average Effects					
	Statistic	Volatility σ	Factor Loading β	Systematic Share R^2	Unshared Risk ρ
Before	Mean	0.117	1.0	0.515	0.717
	Stdev	0.037	0.367	0.255	0.093
After	Mean	0.113	0.945	0.498	0.701
Difference		-0.004	-0.055	-0.017	-0.016
Fraction of Stdev		0.108	0.151	0.066	0.169
Panel B: Conditional Effects					
	Statistic	Volatility σ	Factor Loading β	Systematic Share R^2	Unshared Risk ρ
Core					
Before	Mean	0.085	0.974	0.511	0.701
	Stdev	0.028	0.434	0.257	0.12
After	Mean	0.082	0.939	0.499	0.689
Difference		-0.002	-0.035	-0.012	-0.012
Fraction of Stdev		0.076	0.081	0.048	0.101
Peripheral					
Before	Mean	0.146	1.004	0.51	0.706
	Stdev	0.032	0.187	0.174	0.061
After	Mean	0.14	0.95	0.485	0.688
Difference		-0.006	-0.054	-0.025	-0.018
Fraction of Stdev		0.191	0.29	0.141	0.294

This table shows the results of a counterfactual simulation that changes trade costs to match a 50 percent growth in imports across two countries. We do this for each bilateral pair, $N(N-1)/2$ times, and track the average change across these pairs as After. We compare these to Before where we report the mean and standard deviation across countries. Core and peripheral countries are defined in the relation to median($\bar{\varphi}_j$) where $\bar{\varphi}_j = \sum_{i \neq j} \varphi_{ij} / (N-1)$ as defined in (30).

Table 12
A US-China Trade War

Panel A: Effects on United States and China						
Variable	United States		China			
	Before	After	Before	After		
Net Trade Cost, $\tau - 1$	0.947	1.42	1.242	1.862		
Inward Resistance, P	0.99	0.991	1.062	1.054		
Outward Resistance, Π	1.09	1.09	1.197	1.209		
Import Share, ω	0.027	0.012	0.041	0.014		
Unshared Risk, ρ	0.724	0.785	0.724	0.785		
Panel B: Effects on Peripheral Countries, in Percent Change						
	Canada	Mexico	UK	Korea	Malaysia	Singapore
<i>Trade Variables</i>						
P_j	-0.44	-0.6	-0.85	-0.9	-0.88	-0.9
Π_j	0.04	0.22	0.58	0.42	0.32	0.12
$\omega_{USA,j}$	-1.06	-2.03	-1.05	0.48	-0.15	-0.65
$\omega_{CHN,j}$	19.32	19.2	11.65	8.58	8.4	9.1
$\omega_{i,USA}$	2.14	1.87	3.73	6.28	5.81	4.72
$\omega_{i,CHN}$	11.32	7.72	4.34	2.96	2.89	2.92
<i>Risk Measures</i>						
$\bar{\sigma}_j$	2.03	0.89	0.17	0.1	0.46	0.6
$R^2_{i,USA}$	13.01	12.2	0.94	2.85	0.99	0.1
$R^2_{i,CHN}$	29.88	28.97	4.37	13.31	5.89	2.79
$\rho_{i,USA}$	0.36	-1.08	-0.16	3.62	3.12	2.72
$\rho_{i,CHN}$	8.13	5.75	0.24	5.55	6.88	6.7

This table shows the results of a counterfactual US-China trade war where bilateral trade costs between the United States and China are raised by 50 percent. In Panel A, the values of net trade costs, import shares, and unshared risks are with respect to the other country. In Panel B, average bilateral volatility is $\bar{\sigma}_j = \sum_{i \neq j} \sigma_{ij} / (N-1)$

each country's output, measures of resistance are largely unchanged. Thus their size insulates them from damage. Unsurprisingly, risk-sharing has worsened across the pair.

Panel B reports the effects in percent change on countries peripheral to the US—Canada, Mexico, and the United Kingdom—and China—Korea, Malaysia, and Singapore. The trade war scales up the outward resistances of these trading partners, lowering their output prices. Because most consumption is from domestic sources, their inward price index, on net, falls. The overall effect on the price level is small across countries and does not exceed a one percent change.

Canadian, Mexican, and British exports to China, $\omega_{i,CHN}$, and Chinese imports into these countries, $\omega_{CHN,j}$, expand to replace the void left by the US. A similar story plays out for the three Asian peripherals: because their output price has fallen, the US and China imports more from them. Turning to risk, the average volatility of the six countries' currencies rise with a concomitant growth in their systematic risk: American and Chinese currency investors in these six countries see higher R -squareds and degrees of unshared risk on average.

Overall, the US-China trade war does not have a material effect on its initiators. Instead, the peripheral countries that depend on these giants bear collateral damage and the brunt of the war. Because of the trade war of two, risk-sharing across all has worsened.

5. Conclusion

We provide direct evidence of a strong, temporal link between fundamentals and currency prices by refocusing our empirical work towards the exchange rate factor structure and away from simple changes in exchange rates. Key to uncovering this connection in our paper is the use of trade deals as an instrument for trade.

The typical deal raises bilateral imports across both countries by 50 percent over five years and leads to a reduction in all measures of exchange rate systematic risk: in particular, the share of systematic risk and our new measure of unshared risk fall by, generally, a third relative to the standard deviation across countries.

We use a structural model that generates a gravity equation to estimate trade costs and use it to replicate our first stage and see where the benefits of trade deals concentrate. In this exercise, we find that peripheral countries, which are often poorer, disproportionately benefit from trade deals through significant reductions in the systematic risk of their currencies. Our counterfactual of a severe US-China trade war corroborates this lesson by showing pronounced effects on peripheral countries.

Overall, we provide new evidence that the theatre of political economy and trade negotiations has a direct impact on the risk faced by currency investors. Our work guides future research in international macroeconomics that emphasizes the roles played by interest parity and currency risk in shaping firms' investment and financing decisions.

Declaration of Competing Interest

I have read and agree to the Journal of Financial Economics' Submission Guidelines and Conflict of Interest Disclosure Policy. I hereby declare that I have no relevant or material financial interests that relate to the research described in this paper. All sources of research funding have been explicitly acknowledged in the paper.

Data availability

Data is available at 10.17632/v93hhmtkj8.1

Appendix A. Empirical Appendix

A1. Exchange rate specification for foundation of bilateral risk measures

Rather than assuming the form of SDFs as in Section 1.1, we can instead specify the dynamics of exchange rates and from there derive the risk measures of exchange rates. We write $\Delta s_{i\$,t+1}$ as the log change from t to $t+1$ in the price of currency i in units of the US dollar and, as before, expose the currency to country-specific and global shocks,

$$\Delta \mathbf{E}_t[\Delta s_{i\$,t+1}] = \delta_i u_{i,t+1} + \varphi_i u_{g,t+1} - \delta_\$ u_{\$,t+1} - \varphi_\$ u_{g,t+1}, \quad (\text{A.1})$$

where the notation is similar to those in the paper.

We then define the log change of i and j 's bilateral exchange rate as $\Delta s_{ij,t+1} = \Delta s_{i\$,t+1} - \Delta s_{j\$,t+1}$. Applying the innovations operator to the rate thus gives

$$\Delta \mathbf{E}_t[\Delta s_{ij,t+1}] = \delta_i u_{i,t+1} - \delta_j u_{j,t+1} + (\varphi_i - \varphi_j) u_{g,t+1}, \quad (\text{A.2})$$

which is identical to (2). All calculations for and definitions of risk then mirror those in the paper. And we similarly define base factors as the cross-sectional average currency return, $\Delta base_{j,t+1} = \frac{1}{N} \sum_{i=1}^N \Delta s_{ij,t+1}$.

We start with the reference currency, the U.S. dollar base factor:

$$\Delta \mathbf{E}_t[\Delta base_{\$,t+1}] = \frac{1}{N} \sum_{i=1}^N \Delta \mathbf{E}_t[\Delta s_{i\$,t+1}] = \frac{1}{N} \sum_{i=1}^N \Delta \mathbf{E}_t[s_{i,t+1}] \quad (\text{A.3})$$

$$= \frac{1}{N} \sum_{i=1}^N \delta_i u_{i,t+1} + \frac{1}{N} \sum_{i=1}^N \varphi_i u_{g,t+1} - \delta_\$ u_{\$,t+1} - \varphi_\$ u_{g,t+1}, \quad (\text{A.4})$$

$$= (\bar{\varphi} - \varphi_\$) u_{g,t+1} - \delta_\$ u_{\$,t+1} \quad (\text{A.5})$$

Then the other currency base factors are defined relatively to the dollar base factor:

$$\begin{aligned} \Delta \mathbf{E}_t[\Delta base_{j,t+1}] &= \Delta \mathbf{E}_t \frac{1}{N} \sum_{i=1}^N \Delta s_{ij,t+1} \\ &= \Delta \mathbf{E}_t \frac{1}{N} \sum_{i=1}^N s_{i,t+1} - \Delta \mathbf{E}_t s_{j,t+1} \end{aligned} \quad (\text{A.6})$$

$$= \Delta \mathbf{E}_t[\Delta base_{\$,t+1}] - \Delta \mathbf{E}_t[s_{j,t+1}], \quad (\text{A.7})$$

$$= (\bar{\varphi} - \varphi_j)u_{g,t+1} - \delta_j u_{j,t+1}. \quad (\text{A.8})$$

Similarly the regression of bilateral exchange rates between i and j on the base factor for currency j (equation (4)) yield the same coefficients:

$$\beta_{ij} = \frac{\text{Cov}(\Delta s_{ij,t+1}, \Delta \text{base}_{j,t+1})}{\text{Var}(\Delta \text{base}_{j,t+1})} = \frac{\delta_j^2 + (\bar{\varphi} - \varphi_j)(\varphi_i - \varphi_j)}{\delta_j^2 + (\bar{\varphi} - \varphi_j)^2} \quad (\text{A.9})$$

$$R_{ij}^2 = \beta_{ij}^2 \frac{\text{Var}(\Delta \text{base}_{j,t+1})}{\text{Var}(\Delta s_{ij,t+1})} = \beta_{ij}^2 \times \frac{\delta_j^2 + (\bar{\varphi} - \varphi_j)^2}{\delta_i^2 + \delta_j^2 + (\varphi_i - \varphi_j)^2}. \quad (\text{A.10})$$

While the measure of unshared risk ρ_{ij} is defined based on the a process for SDF, we find the same expression when starting directly from exchange rates:

$$\begin{aligned} \rho_{ij} &= \frac{\text{Var}(\Delta s_{ij,t+1})}{\text{Var}(\Delta \text{base}_{i,t+1}) + \text{Var}(\Delta \text{base}_{j,t+1})} \\ &= \frac{\delta_i^2 + \delta_j^2 + (\varphi_i - \varphi_j)^2}{\delta_i^2 + (\bar{\varphi} - \varphi_i)^2 + \delta_j^2 + (\bar{\varphi} - \varphi_j)^2}. \end{aligned} \quad (\text{A.11})$$

We can also rewrite it as a function of exchange rate directly:

$$\rho_{ij} = \frac{\text{Var}(\Delta s_{ij,t+1})}{\text{Var} s_{i,t+1} + \text{Var} s_{j,t+1} + 2(\bar{\varphi} - \varphi_s)(2\bar{\varphi} - \varphi_i - \varphi_j) - 2\text{Var}(\Delta \text{base}_{s,t+1})}. \quad (\text{A.12})$$

A2. Constructing exchange rate risk measures from the principal component decomposition

We estimate the principal components of the cross-section of base returns to decompose the base returns into a systematic and an idiosyncratic component:

$$\Delta \text{base}_{j,t} = \underbrace{\sum_{l=1}^6 \beta_j^{(l)} \text{PC}_{l,t}}_{\text{Systematic Component}} + \underbrace{\sum_{m>6} \beta_j^{(m)} \text{PC}_{m,t}}_{\text{Idiosyncratic Component}}$$

We therefore recover a time-series of the principal components to decompose the base factors into their systematic component and their idiosyncratic or country-level component. Our choice of six components is driven by the fraction of the variance explained; in our main specification we stop at the last component which explains at least two thirds of the variance, the sixth component.

The projection of the base factors on the first six principal components corresponds to the systematic base factors ($\Delta \text{base}_{j,t}^{\text{sys}}$) while the residual is the idiosyncratic base factor ($\Delta \text{base}_{j,t}^{\text{idio}}$). We also project currency returns onto the first six principal components to obtain the systematic and idiosyncratic components; we estimate the following cross-sectional regression at each point in time to decompose the bilateral currency returns:

$$\Delta s_{ij,t} = \underbrace{\sum_{l=1}^6 \gamma_{ij,t}^{(l)} \text{PC}_{l,t}}_{\text{Systematic Component}} + \varepsilon_{ij,t}$$

The first component corresponds to the systematic component of bilateral exchange rates which we denote $\Delta s_{ij,t}^{\text{sys}}$ and the second component, the residual of the regression, is the idiosyncratic component which we denote $\Delta s_{ij,t}^{\text{idio}}$. With the two components of the base factor and the two components of bilateral exchange rates we use equations (5) to (8) to reestimate the four risk-measures which we use as the dependent variables for Tables 7 and 8 (systematic risk) and Tables A.7 and A.8 (idiosyncratic):

$$(\sigma_{ij}^{\text{sys}})^2 = \text{Var}(\Delta s_{ij,t}^{\text{sys}}) \quad (\sigma_{ij}^{\text{idio}})^2 = \text{Var}(\Delta s_{ij,t}^{\text{idio}}),$$

$$\beta_{ij}^{\text{sys}} = \frac{\text{Cov}(\Delta s_{ij,t}^{\text{sys}}, \Delta \text{base}_{j,t}^{\text{sys}})}{\text{Var}(\Delta \text{base}_{j,t}^{\text{sys}})},$$

$$\beta_{ij}^{\text{idio}} = \frac{\text{Cov}(\Delta s_{ij,t}^{\text{idio}}, \Delta \text{base}_{j,t}^{\text{idio}})}{\text{Var}(\Delta \text{base}_{j,t}^{\text{idio}})},$$

$$(R_{ij}^{\text{sys}})^2 = (\beta_{ij}^{\text{sys}})^2 \frac{\text{Var}(\Delta \text{base}_{j,t}^{\text{sys}})}{\text{Var}(\Delta s_{ij,t}^{\text{sys}})},$$

$$(R_{ij}^{\text{idio}})^2 = (\beta_{ij}^{\text{idio}})^2 \frac{\text{Var}(\Delta \text{base}_{j,t}^{\text{idio}})}{\text{Var}(\Delta s_{ij,t}^{\text{idio}})},$$

$$\rho_{ij}^{\text{sys}} = \frac{\text{Var}(\Delta s_{ij,t}^{\text{sys}})}{\text{Var}(\Delta \text{base}_{i,t}^{\text{sys}}) + \text{Var}(\Delta \text{base}_{j,t}^{\text{sys}})},$$

$$\rho_{ij}^{\text{idio}} = \frac{\text{Var}(\Delta s_{ij,t}^{\text{idio}})}{\text{Var}(\Delta \text{base}_{i,t}^{\text{idio}}) + \text{Var}(\Delta \text{base}_{j,t}^{\text{idio}})}.$$

Appendix B. Model Appendix

B1. Financial autarky

The current value Lagrangian for country j at a point in time is

$$\begin{aligned} \mathcal{L}_j &= u(C_j) + \lambda_j^{\text{RC}} \left(\left(\sum_{i=1}^N q_{ij}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} - C_j \right) \\ &+ \lambda_j^{\text{BC}} \left(w_j L_j - \sum_{i=1}^N p_{ij} q_{ij} \right) \end{aligned} \quad (\text{B.1})$$

The system of first-order conditions is

$$C_j : \quad u'(C_j) = \lambda_j^{\text{RC}} \quad (\text{B.2})$$

$$q_{ij} : \quad \lambda_j^{\text{BC}} p_{ij} = \lambda_j^{\text{RC}} C_j^{\frac{1}{\eta}} q_{ij}^{-\frac{1}{\eta}} \quad (\text{B.3})$$

Equation (C.3) can be multiplied by q_{ij} and integrated to give the expression

$$\lambda_j^{RC} / \lambda_j^{BC} = \frac{\sum_{i=1}^N p_{ij} q_{ij}}{C_j} = \frac{w_j L_j}{C_j}, \quad (\text{B.4})$$

where the second equality follows from $\sum_{i=1}^N p_{ij} q_{ij} = w_j L_j$. Substituting (C.4) into (C.3) and integrating gives the CES demand equation in (21).

B2. Complete markets

The current value Lagrangian of the planner's problem is

$$\begin{aligned} \mathcal{L} = & \sum_j \lambda_j u(C_j) + \lambda_j^{RC} \left(\left(\sum_{i=1}^N q_{ij}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} - C_j \right) \\ & + \lambda_j^{GRC} \left(\sum_{j=1}^N \tau_{jk} q_{jk} - z_j l_{Tj} \right) \end{aligned} \quad (\text{B.5})$$

The first-order conditions are

$$C_j : \quad \lambda_j u'(C_j) = \lambda_j^{RC} \quad (\text{B.6})$$

$$q_{ij} : \quad \lambda_i^{GRC} \tau_{ij} = \lambda_j^{RC} C_j^{\frac{1}{\eta}} q_{ij}^{-\frac{1}{\eta}} \quad (\text{B.7})$$

Comparing these conditions to autarky's, it is evident under the normalization $\lambda_j^{BC} = 1/\lambda_j$ that the equilibrium prices of j 's tradable good, p_{jj} , are equal to the shadow values of j 's global resource constraint, λ_j^{GRC} .

Appendix C. Appendix Tables

A.1–A.16

Table A.1
Summary Statistics for Developed and Emerging Countries

	Observations	Mean	Median	Std. Dev.
<i>Developed</i>				
Base Loadings, β	26,794	1.036	1.000	0.917
Share of Systematic Risk, R^2	26,766	0.471	0.465	0.295
Volatility, σ	26,794	0.130	0.114	0.105
Unshared Risk, ρ	26,794	0.916	0.965	0.301
Imports (billions of USD)	45,400	8.870	1.084	36.36
Imports, five-year difference (billions of USD)	39,798	2.137	0.210	11.56
Imports ten-year difference (billions of USD)	34,809	4.733	0.614	20.33
Imports, as a share of GDP	40,672	0.029	0.006	0.102
CUMECU	54,050	0.147	0.000	0.354
FTA	54,050	0.145	0.000	0.352
TWPTA	54,050	0.022	0.000	0.145
OWPTA	54,050	0.035	0.000	0.184
<i>Emerging</i>				
Base Loadings, β	28,986	0.989	0.992	0.540
Share of Systematic Risk, R^2	28,976	0.521	0.525	0.315
Volatility, σ	28,986	0.157	0.126	0.136
Unshared Risk, ρ	28,986	0.907	0.965	0.317
Imports (billions of USD)	48,020	3.584	0.348	16.92
Imports, five-year difference (billions of USD)	41,878	1.183	0.085	6.649
Imports ten-year difference (billions of USD)	36,369	2.664	0.260	11.83
Imports, as a share of GDP	36,976	0.023	0.005	0.060
CUMECU	58,750	0.034	0.000	0.181
FTA	58,750	0.084	0.000	0.278
TWPTA	58,750	0.048	0.000	0.214
OWPTA	58,750	0.263	0.000	0.440

The table reports summary statistics of exchange rates and trade at the country level.

Base loadings and the R -squared are estimated from the regression in (4).

The import shares are estimated from IMF and CEPII data over the period 1973–2019.

Trade deals are divided in four different levels of economic integration as in Baier and Bergstrand (2007): customs unions, common markets, and economic unions (CUMECU), free trade arrangement (FTA), one-way and two-way preferential trade agreement (OWPTA and TWPTA).

Table A.2
Summary Statistics of the Systematic and Idiosyncratic Components of Exchange Rate Risk

	Observations	Mean	Median	Std. Dev.
<i>Systematic Risk Measures</i>				
Base Loadings, β	53,799	1.033	0.991	1.195
Share of Systematic Risk, R^2	53,785	0.494	0.500	0.340
Volatility, σ	53,799	0.120	0.096	0.137
Unshared Risk, ρ	53,681	0.884	0.958	0.424
<i>Idiosyncratic Risk Measures</i>				
Base Loadings, β	53,799	1.009	0.986	0.359
Share of Systematic Risk, R^2	53,785	0.518	0.513	0.345
Volatility, σ	53,799	0.082	0.074	0.042
Unshared Risk, ρ	53,681	0.963	0.957	0.211

The table reports summary statistics of exchange rates and trade at the country level. The risk measures have been separated into systematic and idiosyncratic components via a principal components analysis (see Section 1.5). Base loadings and the R-squared are estimated from the regression in (4).

Table A.3
Time Effects of Trade Deal Change. Staggered Difference-in-Differences with Country-Pair Fixed Effects

Timing	Log Import		Volatility, σ		Unshared Risk, ρ		R-squared		Base Loadings, β	
	β	s.e.	β	s.e.	β	s.e.	β	s.e.	β	s.e.
-10	-0.019	(0.041)	0.001	(0.002)	-0.004	(0.016)	0.002	(0.013)	0.101	(0.077)
-9	-0.032	(0.039)	-0.003	(0.002)	-0.025	(0.017)	-0.008	(0.014)	0.059	(0.065)
-8	-0.046	(0.037)	-0.001	(0.002)	-0.011	(0.016)	-0.003	(0.013)	0.066	(0.055)
-7	-0.052	(0.036)	-0.001	(0.002)	-0.023	(0.016)	-0.009	(0.014)	0.057	(0.051)
-6	-0.045	(0.035)	-0.001	(0.002)	0.004	(0.018)	0.009	(0.016)	0.077*	(0.046)
-5	-0.065**	(0.032)	-0.001	(0.002)	-0.009	(0.015)	-0.002	(0.013)	0.052	(0.043)
-4	-0.066**	(0.030)	-0.000	(0.002)	-0.006	(0.013)	-0.000	(0.011)	0.044	(0.033)
-3	-0.060**	(0.026)	0.002	(0.001)	0.007	(0.012)	0.006	(0.010)	0.056**	(0.027)
-2	-0.046**	(0.021)	-0.000	(0.001)	-0.001	(0.010)	0.004	(0.009)	0.037*	(0.020)
-1	-0.025	(0.018)	0.000	(0.001)	0.010	(0.008)	0.009	(0.007)	0.026**	(0.013)
1	0.038**	(0.017)	-0.001	(0.001)	-0.014***	(0.005)	-0.006	(0.004)	-0.007	(0.011)
2	0.057***	(0.019)	-0.001	(0.001)	-0.011	(0.008)	-0.004	(0.007)	-0.010	(0.012)
3	0.075***	(0.025)	-0.000	(0.001)	-0.008	(0.010)	-0.004	(0.009)	-0.023	(0.014)
4	0.099***	(0.024)	-0.001	(0.001)	-0.011	(0.012)	-0.007	(0.010)	-0.039**	(0.019)
5	0.118***	(0.027)	0.000	(0.001)	-0.003	(0.012)	-0.002	(0.010)	-0.026	(0.021)
6	0.154***	(0.030)	-0.001	(0.001)	-0.012	(0.014)	-0.008	(0.012)	-0.043*	(0.024)
7	0.149***	(0.033)	-0.001	(0.001)	-0.014	(0.014)	-0.007	(0.011)	-0.051**	(0.026)
8	0.163***	(0.034)	-0.005***	(0.002)	-0.072***	(0.015)	-0.038***	(0.012)	-0.117***	(0.028)
9	0.164***	(0.035)	-0.005***	(0.001)	-0.043***	(0.014)	-0.023**	(0.011)	-0.106***	(0.038)
10	0.177***	(0.038)	-0.004**	(0.001)	-0.017	(0.015)	-0.008	(0.012)	-0.074*	(0.039)
Observations	84134		49102		49102		49064		49102	
R-squared	0.947		0.977		0.705		0.824		0.771	

This table reports the coefficient of a staggered difference as in (12). The deal dummy equals one if countries i and j share either an FTA or a CUCMECU and zero otherwise. The set of control variables X_{ijt} includes country-pair fixed effects, time fixed effects, and the deal dummy. All time coefficients are deviations in y_{ijt} from the exact year of the trade deal change.

Table A.4
Staggered Difference-in-Differences with Aggregation from Callaway and Sant'Anna (2019)

Timing	Log Import		Volatility, σ		Unshared Risk, ρ		R-squared		Base Loadings, β	
	β	s.e.	β	s.e.	β	s.e.	β	s.e.	β	s.e.
-10	-0.034	(0.030)	-0.003	(0.004)	-0.044***	(0.009)	-0.021	(0.017)	-0.061***	(0.020)
-9	-0.038	(0.038)	-0.000	(0.002)	-0.022***	(0.005)	-0.013*	(0.009)	-0.042***	(0.013)
-8	0.095***	(0.027)	-0.001	(0.003)	0.011*	(0.008)	-0.001	(0.016)	0.004	(0.016)
-7	-0.033	(0.034)	0.008*	(0.005)	0.020	(0.017)	0.009	(0.016)	0.025*	(0.019)
-6	0.012	(0.048)	0.006**	(0.004)	-0.012	(0.011)	-0.006	(0.011)	0.003	(0.018)
-5	-0.054***	(0.023)	-0.001	(0.002)	-0.019**	(0.008)	-0.014*	(0.009)	-0.033***	(0.011)
-4	0.092***	(0.025)	-0.004**	(0.002)	0.006	(0.076)	0.000	(0.046)	0.036	(0.095)
-3	0.165**	(0.074)	0.000	(0.004)	-0.005	(0.020)	-0.004	(0.057)	-0.016	(0.025)
-2	-0.242***	(0.069)	-0.028***	(0.008)	0.039***	(0.015)	0.023	(0.038)	0.016	(0.087)
-1	-0.016	(0.022)	0.000	(0.002)	0.024**	(0.010)	0.008	(0.012)	-0.015	(0.036)
1	0.311***	(0.050)	-0.025***	(0.008)	-0.083***	(0.022)	-0.046	(0.043)	-0.117***	(0.046)
2	0.324***	(0.051)	-0.050***	(0.008)	-0.153***	(0.026)	-0.087	(0.087)	-0.189***	(0.055)
3	0.316***	(0.059)	-0.054***	(0.009)	-0.159***	(0.027)	-0.093	(0.096)	-0.214***	(0.050)
4	0.200***	(0.069)	-0.061***	(0.007)	-0.152***	(0.021)	-0.090	(0.073)	-0.230***	(0.038)
5	0.108*	(0.074)	-0.069***	(0.007)	-0.196***	(0.022)	-0.126**	(0.072)	-0.288***	(0.051)
6	0.271***	(0.062)	-0.071***	(0.007)	-0.218***	(0.025)	-0.132**	(0.068)	-0.307***	(0.048)
7	0.255***	(0.064)	-0.059***	(0.007)	-0.160***	(0.022)	-0.095*	(0.065)	-0.246***	(0.044)
8	0.526***	(0.073)	-0.059***	(0.007)	-0.189***	(0.024)	-0.110***	(0.038)	-0.281***	(0.048)
9	0.418***	(0.112)	-0.060***	(0.009)	-0.203***	(0.028)	-0.117***	(0.043)	-0.294***	(0.051)
10	0.541***	(0.122)	-0.060***	(0.008)	-0.213***	(0.027)	-0.122***	(0.044)	-0.303***	(0.050)

This table reports the coefficient of a staggered difference as expressed in the specification (12). The deal dummy equals one if countries i and j share either an FTA or a CUCMECU and zero otherwise. The set of controls include country-pair and year fixed effects. To aggregate the difference-in-differences estimate across treated unit, we follow the implementation of Callaway and Sant'Anna (2019) and report coefficients and standard errors from a dynamic aggregation of the average treatment effect.

Table A.5
OLS Specification of Trade on Exchange Rate Risk

	Panel A: Volatility, σ					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5}$ log Import	0.010 (0.022)	0.009 (0.021)	0.007 (0.021)	0.003 (0.019)	-0.004 (0.016)	-0.005 (0.016)
$\Delta_{t-5,t-10}$ log Import	0.004 (0.011)	0.002 (0.011)	0.002 (0.011)	0.020* (0.012)	0.018* (0.009)	0.018* (0.010)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,808	33,808	33,047	20,223	20,223	19,581
	Panel B: Unshared Risk, ρ					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5}$ log Import	0.048 (0.194)	-0.139 (0.190)	-0.163 (0.190)	-0.054 (0.195)	-0.146 (0.159)	-0.176 (0.158)
$\Delta_{t-5,t-10}$ log Import	0.008 (0.121)	-0.098 (0.117)	-0.099 (0.116)	0.156 (0.123)	0.112 (0.102)	0.107 (0.097)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,808	33,808	33,047	20,223	20,223	19,581

Table A.5 reports regressions of volatility (Panel A) and unshared risk (Panel B) on the change in imports. Columns (1) through (3) report the five-year change in volatility and columns (4) through (6) the ten-year change. All dependent variables were multiplied by 100 because of the small magnitude of the coefficients. Controls include pre-trends, country-year fixed effects for both countries, gravity variables in columns (1) and (4), and country pair fixed effects in columns (2), (3), (5), and (6). In columns (3) and (6) we remove pegged currencies. We report standard errors clustered at country-pair level in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent level.

Table A.6
OLS Specification of Trade on Common Factor Regressions

	Panel A. Base Loading, β					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \log \text{Import}$	0.619 (0.673)	0.504 (0.617)	0.494 (0.626)	0.441 (0.567)	0.275 (0.527)	0.246 (0.533)
$\Delta_{t-5,t-10} \log \text{Import}$	0.342 (0.257)	0.333 (0.265)	0.349 (0.267)	0.175 (0.178)	0.249 (0.153)	0.245 (0.155)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,808	33,808	33,047	20,223	20,223	19,581
	Panel B. Share of Systematic Risk, R^2					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \log \text{Import}$	0.079 (0.182)	0.009 (0.175)	0.000 (0.176)	-0.206 (0.150)	-0.253* (0.135)	-0.279** (0.133)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.088 (0.121)	-0.133 (0.114)	-0.137 (0.112)	0.092 (0.102)	0.112 (0.084)	0.137 (0.092)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,776	33,776	33,017	20,197	20,197	19,557

This table shows the instrumental variable regressions of the slope coefficient (Panel A) and R-squared (Panel B) from [equation \(4\)](#) on the change in imports. Columns (1) through (3) report the five-year change in volatility and columns (4) through (6) the ten-year change. All dependent variables were multiplied by 100 because of the small magnitude of the coefficients. Controls include pre-trends, country-year fixed effects for both countries, gravity variables in columns (1) and (4), and country pair fixed effects in columns (2), (3), (5), (6). In columns (3) and (6) we remove pegged currencies. We report standard errors clustered at country-pair level in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent level.

Table A.7
Time Effects of Trade Deal Change on the Idiosyncratic Risk Measures. Staggered Difference-in-Difference Regression.

Timing	Log Import		Volatility, σ		Unshared Risk, ρ		R-squared		Base Loadings, β	
	β	s.e.	β	s.e.	β	s.e.	β	s.e.	β	s.e.
-10	-0.072	(0.052)	0.002	(0.002)	0.374	(0.315)	-0.001	(0.001)	0.003	(0.004)
-9	-0.069	(0.050)	-0.000	(0.002)	0.365**	(0.144)	0.000	(0.001)	0.004	(0.004)
-8	-0.099**	(0.049)	-0.001	(0.002)	0.274**	(0.129)	-0.001	(0.001)	0.000	(0.006)
-7	-0.126***	(0.043)	0.002	(0.002)	0.197	(0.138)	0.000	(0.000)	0.001	(0.004)
-6	-0.086**	(0.042)	0.001	(0.002)	0.205*	(0.117)	0.001*	(0.001)	0.005	(0.004)
-5	-0.064*	(0.038)	-0.003	(0.004)	0.043	(0.131)	0.000	(0.000)	0.003	(0.003)
-4	-0.068**	(0.034)	0.001	(0.003)	0.131	(0.125)	-0.000	(0.001)	-0.001	(0.005)
-3	-0.069**	(0.029)	0.001	(0.003)	0.077	(0.101)	0.000	(0.001)	0.012**	(0.006)
-2	-0.023	(0.025)	0.004*	(0.002)	0.200**	(0.082)	0.003**	(0.002)	0.005	(0.007)
-1	0.011	(0.018)	0.004***	(0.001)	0.209***	(0.065)	0.000	(0.001)	-0.006	(0.005)
1	0.039**	(0.019)	-0.000	(0.001)	-0.059**	(0.028)	0.001***	(0.000)	-0.000	(0.003)
2	0.067***	(0.025)	0.000	(0.001)	-0.059	(0.044)	0.000	(0.000)	0.001	(0.003)
3	0.085**	(0.034)	-0.001	(0.001)	-0.135**	(0.054)	0.000*	(0.000)	0.001	(0.003)
4	0.104***	(0.035)	-0.002*	(0.001)	-0.069	(0.064)	0.000	(0.000)	0.003	(0.003)
5	0.114***	(0.037)	-0.003*	(0.002)	-0.144**	(0.068)	0.000	(0.000)	0.001	(0.003)
6	0.124***	(0.044)	-0.003*	(0.002)	-0.278***	(0.077)	0.000	(0.000)	-0.000	(0.003)
7	0.207***	(0.051)	-0.005***	(0.002)	-0.305***	(0.081)	0.000	(0.000)	0.000	(0.003)
8	0.216***	(0.055)	-0.014***	(0.002)	-0.879***	(0.111)	0.000	(0.000)	-0.008**	(0.004)
9	0.200***	(0.057)	-0.013***	(0.002)	-1.363***	(0.437)	0.002***	(0.000)	0.001	(0.003)
10	0.243***	(0.061)	-0.012***	(0.002)	-0.760***	(0.155)	0.000*	(0.000)	0.004	(0.003)
Observations	44924		48471		48353		48457		48471	
R-squared	0.841		0.938		0.942		0.790		0.445	

This table reports coefficients from a staggered difference-in-difference regression as in [\(12\)](#). The risk measures are the residuals from a principal components analysis (see [Section 1.5](#)). The deal dummy equals one if countries i and j share either an FTA or a CUCMECU and zero otherwise. The set of control variables X_{ijt} includes gravity variables, time fixed effects, and the deal dummy. All time coefficients are deviations in y_{ijt} from the exact year of the trade deal change.

Table A.8

Second Stage: Effect of Trade on Common Factor Regressions for the Idiosyncratic Risk Measures

	Panel A					
	Volatility, σ			Unshared Risk, ρ		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.017** (0.009)	-0.021** (0.011)	-0.021** (0.011)	-0.988** (0.499)	-1.024* (0.588)	-0.997* (0.583)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.002 (0.001)	-0.004* (0.003)	-0.004* (0.003)	-0.145* (0.080)	-0.214 (0.138)	-0.206 (0.136)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	18,312	18,312	17,959	18,264	18,264	17,911
First stage F-statistic	5.97	5.11	5.28	5.94	5.07	5.24

	Panel B					
	Base Loading, β			Share of Systematic Risk, R^2		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	0.009 (0.011)	0.006 (0.013)	0.006 (0.013)	0.001* (0.000)	0.001 (0.001)	0.001 (0.001)
$\Delta_{t-5,t-10} \log \text{Import}$	0.001 (0.002)	0.001 (0.003)	0.001 (0.003)	0.000* (0.000)	0.000 (0.000)	0.000* (0.000)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	18,312	18,312	17,959	18,312	18,312	17,959
First stage F-statistic	5.97	5.11	5.28	5.97	5.11	5.28

This table shows the instrumental variable regressions of five-year changes in volatility and unshared risk (Panel A) and base loadings and R -squared (Panel B) from [equation \(4\)](#) on the change in imports, instrumented using trade deals. The risk measures are the residuals from a principal components analysis (see [Section 1.5](#)). Controls include pre-trends, country-year fixed effects for both countries, gravity variables in columns (1) and (4), and country-pair fixed effects in columns (2), (3), (5), and (6). In columns (3) and (6) we remove pegged currencies. We report standard errors clustered at the country-pair level in parentheses and the first-stage Cragg-Donald F-statistics. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

Table A.9

Second Stage: Effect of Trade on Exchange Rate Risk. Peripheral Countries

	Panel A: Volatility, σ					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.029** (0.012)	-0.044** (0.020)	-0.028** (0.012)	-0.021*** (0.008)	-0.015* (0.008)	-0.011 (0.007)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.008** (0.003)	-0.014** (0.007)	-0.009** (0.004)	0.002** (0.001)	-0.000 (0.000)	-0.000 (0.000)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	16,302	16,302	15,876	10,204	10,204	9,834
First stage F-statistic	6.79	3.87	6.73	8.52	6.06	5.78

(continued on next page)

Table A.9
(continued)

	Panel B: Unshared Risk, ρ					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.408*** (0.150)	-0.576** (0.252)	-0.361*** (0.132)	-0.288*** (0.094)	-0.263** (0.108)	-0.186** (0.089)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.114*** (0.044)	-0.179** (0.082)	-0.113*** (0.043)	0.024** (0.010)	-0.006 (0.005)	-0.005 (0.004)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	16,302	16,302	15,876	10,204	10,204	9,834
First stage F-statistic	6.79	3.87	6.73	8.52	6.06	5.78

This table shows the instrumental variable regressions of volatility (Panel A) and unshared risk (Panel B) on the change in imports, instrumented using trade deals. Sample is restricted to base countries belonging to bottom 50 percent of countries, as defined by the centrality measure in [equation \(15\)](#). Columns (1) through (3) report the five-year change in volatility and columns (4) through (6) the 10-year change. Controls include pre-trends, country-year fixed effects for both countries, gravity variables in Columns (1) and (4), and country pair fixed effects in columns (2), (3), (5), and (6). In columns (3) and (6) we remove pegged currencies. We report standard errors clustered at the country-pair level in parentheses and the first-stage Cragg-Donald F-statistics. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

Table A.10
Second Stage: Effect of Trade on Exchange Rate Risk. Peripheral Countries

	Panel A. Base Loading, β					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.370** (0.147)	-0.512** (0.234)	-0.332** (0.139)	-0.270*** (0.094)	-0.295** (0.121)	-0.242** (0.110)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.103** (0.043)	-0.159** (0.076)	-0.103** (0.045)	0.023** (0.010)	-0.006 (0.006)	-0.005 (0.005)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	16,302	16,302	15,876	10,204	10,204	9,834
First stage F-statistic	6.79	3.87	6.73	8.52	6.06	5.78

	Panel B. Share of Systematic Risk, R^2					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.237*** (0.086)	-0.292** (0.132)	-0.179** (0.072)	-0.179*** (0.056)	-0.155** (0.065)	-0.108** (0.053)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.067*** (0.025)	-0.092** (0.043)	-0.057** (0.023)	0.014** (0.006)	-0.003 (0.003)	-0.002 (0.003)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	16,298	16,298	15,874	10,202	10,202	9,834
First stage F-statistic	6.79	3.87	6.73	8.53	6.06	5.78

This table shows the instrumental variable regressions of the slope coefficient (Panel A) and R-squared (Panel B) on the change in imports, instrumented using trade deals. Sample is restricted to base countries belonging to bottom 50 percent of countries, as defined by the centrality measure in [equation \(15\)](#). Columns (1) through (3) report the five-year change in volatility and columns (4) through (6) the 10-year change. Controls include pre-trends, country-year fixed effects for both countries, gravity variables in Columns (1) and (4), and country pair fixed effects in columns (2), (3), (5), and (6). In columns (3) and (6) we remove pegged currencies. We report standard errors clustered at the country-pair level in parentheses and the first-stage Cragg-Donald F-statistics. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

Table A.11

Second Stage: Effect of Trade on Exchange Rate Risk. Core Countries

	Panel A: Volatility, σ					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.012 (0.011)	-0.025** (0.012)	-0.023** (0.011)	-0.024*** (0.008)	-0.025*** (0.007)	-0.022*** (0.007)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.001 (0.002)	-0.005* (0.003)	-0.004* (0.002)	0.003** (0.001)	0.000 (0.001)	0.000 (0.001)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	17,475	17,475	17,144	10,019	10,019	9,747
First stage F-statistic	7.09	5.77	5.98	11.7	10.8	11.4
	Panel B: Unshared Risk, ρ					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.224* (0.129)	-0.255* (0.136)	-0.227* (0.130)	-0.450*** (0.128)	-0.359*** (0.111)	-0.307*** (0.101)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.030 (0.022)	-0.048 (0.029)	-0.043 (0.028)	0.054** (0.021)	0.004 (0.015)	0.003 (0.014)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	17,475	17,475	17,144	10,019	10,019	9,747
First stage F-statistic	7.09	5.77	5.98	11.7	10.8	11.4

This table shows the instrumental variable regressions of volatility (Panel A) and unshared risk (Panel B) on the change in imports, instrumented using trade deals. Sample is restricted to base countries belonging to top 50 percent of countries, as defined by the centrality measure in [equation \(15\)](#). Columns (1) through (3) report the five-year change in volatility and columns (4) through (6) the 10-year change. Controls include pre-trends, country-year fixed effects for both countries, gravity variables in Columns (1) and (4), and country pair fixed effects in columns (2), (3), (5), and (6). In columns (3) and (6) we remove pegged currencies. We report standard errors clustered at the country-pair level in parentheses and the first-stage Cragg-Donald F-statistics. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

Table A.12

Second Stage: Effect of Trade on Exchange Rate Risk. Core Countries

	Panel A: Base Loading, β					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.299 (0.222)	-0.326 (0.224)	-0.320 (0.221)	-0.584*** (0.192)	-0.532*** (0.203)	-0.467** (0.195)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.036 (0.035)	-0.056 (0.047)	-0.055 (0.046)	0.069** (0.029)	0.006 (0.023)	0.006 (0.022)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	17,475	17,475	17,144	10,019	10,019	9,747
First stage F-statistic	7.09	5.77	5.98	11.7	10.8	11.4

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Table A.12
(continued)

	Panel B. Share of Systematic Risk, R^2					
	Five-year change			Ten-year change		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \widehat{\log \text{Import}}$	-0.087 (0.088)	-0.145 (0.105)	-0.135 (0.102)	-0.195*** (0.074)	-0.205** (0.080)	-0.184** (0.077)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.012 (0.013)	-0.028 (0.022)	-0.026 (0.021)	0.024** (0.011)	0.003 (0.009)	0.004 (0.009)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	17,447	17,447	17,116	9,995	9,995	9,723
First stage F-statistic	7.06	5.77	5.99	11.7	10.8	11.4

This table shows the instrumental variable regressions of the slope coefficient (Panel A) and R-squared (Panel B) on the change in imports, instrumented using trade deals. Sample is restricted to base countries belonging to top 50 percent of countries, as defined by the centrality measure in equation (15). Columns (1) through (3) report the five-year change in volatility and columns (4) through (6) the 10-year change. Controls include pre-trends, country-year fixed effects for both countries, gravity variables in Columns (1) and (4), and country pair fixed effects in columns (2), (3), (5), and (6). In columns (3) and (6) we remove pegged currencies. We report standard errors clustered at the country-pair level in parentheses and the first-stage Cragg-Donald F-statistics. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

Table A.13
First Stage: Trade and Trade Deals. Only Bilateral Trade Deals

	$\Delta_{t+5,t} \log \text{Import}$			
	(1)	(2)	(3)	(4)
$\Delta_{t,t-5} \text{CUCMECU}$	0.088 (0.098)	0.095 (0.099)	0.117 (0.096)	0.134 (0.096)
$\Delta_{t,t-5} \text{FTA}$	0.031 (0.029)	0.057* (0.030)	0.057* (0.030)	0.069** (0.030)
$\Delta_{t,t-5} \text{TWPTA}$	-0.024 (0.060)	-0.061 (0.064)		
$\Delta_{t,t-5} \text{OWPTA}$	0.105 (0.087)	0.077 (0.097)		
$\Delta_{t-5,t} \log \text{Import}$	-0.217*** (0.029)	-0.246*** (0.027)	-0.246*** (0.027)	-0.249*** (0.028)
Gravity Controls	Y	–	–	–
Country pair f.e.	N	Y	Y	Y
Country-year f.e.	Y	Y	Y	Y
Exchange rate regime	All	All	All	No Pegs
Observations	69,940	69,940	69,940	66,427
R-squared	0.314	0.377	0.377	0.379

This table shows the estimates of a first stage OLS regression of log imports on trade agreement dummies for only bilateral deals. All regressions control for country-time fixed effects. Column (1) controls for gravity variables. In columns (2) through (4), the gravity variables are replaced by country-pair fixed effects. Column (4) also removes pegged currencies. Standard errors are clustered at base-foreign country level. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

Table A.14

First Stage: Trade and Trade Deals. Bilateral Trade Deals Excluding Changes in Anchors

	$\Delta_{t+5,t} \log \text{Import}$			
	(1)	(2)	(3)	(4)
$\Delta_{t,t-5} \text{CUCMECU}$	0.088 (0.098)	0.095 (0.099)	0.117 (0.096)	0.134 (0.096)
$\Delta_{t,t-5} \text{FTA}$	0.030 (0.029)	0.057* (0.031)	0.057* (0.030)	0.069** (0.031)
$\Delta_{t,t-5} \text{TWPTA}$	-0.022 (0.061)	-0.062 (0.065)		
$\Delta_{t,t-5} \text{OWPTA}$	0.105 (0.087)	0.077 (0.097)		
$\Delta_{t-5,t} \log \text{Import}$	-0.217*** (0.029)	-0.246*** (0.027)	-0.246*** (0.027)	-0.249*** (0.028)
Gravity Controls	Y	–	–	–
Country pair f.e.	N	Y	Y	Y
Country-year f.e.	Y	Y	Y	Y
Exchange rate regime	All	All	All	No Pegs
Observations	69,794	69,794	69,794	66,281
R-squared	0.314	0.377	0.377	0.379

This table shows the estimates of a first stage OLS regression of log imports on trade agreement dummies for only bilateral deals that exclude those with changes in anchors. All regressions control for country-time fixed effects. Column (1) controls for gravity variables. In columns (2) through (4), the gravity variables are replaced by country-pair fixed effects. Column (4) also removes pegged currencies. We use the currency anchor classification of Ilzetzi et al. (2019). Standard errors are clustered at base-foreign country level. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

Table A.15

Second Stage: Effect of Trade on Exchange Rate Risk. Only Bilateral Trade Deals

	Panel A					
	Volatility, σ			Unshared Risk, ρ		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \log \widehat{\text{Import}}$	-0.018* (0.010)	-0.031* (0.019)	-0.020 (0.013)	-0.247** (0.118)	-0.479** (0.207)	-0.310** (0.134)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.004* (0.002)	-0.009* (0.005)	-0.005 (0.004)	-0.058** (0.029)	-0.131** (0.058)	-0.085** (0.038)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,777	33,777	33,020	33,777	33,777	33,020
First stage F-statistic	14.2	6.72	8.53	14.2	6.72	8.53

	Panel B					
	Base Loading, β			Share of Systematic Risk, R^2		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \log \widehat{\text{Import}}$	-0.221 (0.144)	0.125 (0.402)	0.102 (0.338)	-0.138* (0.078)	-0.274** (0.134)	-0.170* (0.091)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.050 (0.035)	0.036 (0.109)	0.030 (0.092)	-0.034* (0.019)	-0.076** (0.038)	-0.048* (0.025)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,777	33,777	33,020	33,745	33,745	32,990
First stage F-statistic	14.2	6.72	8.53	14.2	6.7	8.52

This table shows the instrumental variable regressions of the four risk measures on the change in imports, instrumented using trade deals. Only bilateral deals are included. All regressions control for country-time fixed effects. Columns (1) and (4) control for gravity variables. Columns (2) and (5) replace gravity controls with country-pair fixed effects. Columns (3) and (6) also remove pegged currencies. Standard errors are clustered at base-foreign country level. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

Table A.16

Second Stage: Effect of Trade on Exchange Rate Risk. Bilateral Trade Deals Excluding Changes in Anchors

	Panel A					
	Volatility, σ			Unshared Risk, ρ		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \log \widehat{\text{Import}}$	-0.018* (0.010)	-0.031* (0.018)	-0.020 (0.013)	-0.232** (0.115)	-0.458** (0.199)	-0.288** (0.128)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.004* (0.002)	-0.009* (0.005)	-0.005 (0.004)	-0.054* (0.028)	-0.125** (0.056)	-0.079** (0.036)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,667	33,667	32,910	33,667	33,667	32,910
First stage F-statistic	14	6.86	8.79	14	6.86	8.79

	Panel B					
	Base Loading, β			Share of Systematic Risk, R^2		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{t,t-5} \log \widehat{\text{Import}}$	-0.206 (0.143)	0.131 (0.388)	0.112 (0.326)	-0.130* (0.077)	-0.262** (0.129)	-0.158* (0.087)
$\Delta_{t-5,t-10} \log \text{Import}$	-0.046 (0.034)	0.037 (0.106)	0.033 (0.089)	-0.032* (0.019)	-0.072** (0.036)	-0.044* (0.024)
Gravity Controls	Y	–	–	Y	–	–
Country pair f.e.	N	Y	Y	N	Y	Y
Country-year f.e.	Y	Y	Y	Y	Y	Y
Exchange rate regime	All	All	No Pegs	All	All	No pegs
Observations	33,667	33,667	32,910	33,635	33,635	32,880
First stage F-statistic	14	6.86	8.79	14	6.85	8.77

This table shows the instrumental variable regressions of the four risk measures on the change in imports, instrumented using trade deals. Only bilateral deals that do not result in anchor changes are included. All regressions control for country-time fixed effects. Columns (1) and (4) control for gravity variables. Columns (2) and (5) replace gravity controls with country-pair fixed effects. Columns (3) and (6) also remove pegged currencies. Standard errors are clustered at base-foreign country level. ***, **, and * indicate significance at the 1, 5, and 10 percent level, respectively.

C1. Appendix Figures

C2. Financial autarky

The current value Lagrangian for country j at a point in time is

$$\mathcal{L}_j = u(C_j) + \lambda_j^{\text{RC}} \left(\left(\sum_{i=1}^N q_{ij}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} - C_j \right) + \lambda_j^{\text{BC}} \left(w_j L_j - \sum_{i=1}^N p_{ij} q_{ij} \right) \quad (\text{C.1})$$

The system of first-order conditions is

$$C_j : u'(C_j) = \lambda_j^{\text{RC}} \quad (\text{C.2})$$

$$q_{ij} : \lambda_j^{\text{RC}} p_{ij} = \lambda_j^{\text{RC}} C_j^{\frac{1}{\eta}} q_{ij}^{-\frac{1}{\eta}} \quad (\text{C.3})$$

Equation (C.3) can be multiplied by q_{ij} and integrated to give the expression

$$\lambda_j^{\text{RC}} / \lambda_j^{\text{BC}} = \frac{\sum_{i=1}^N p_{ij} q_{ij}}{C_j} = \frac{w_j L_j}{C_j}, \quad (\text{C.4})$$

where the second equality follows from $\sum_{i=1}^N p_{ij} q_{ij} = w_j L_j$. Substituting (C.4) into (C.3) and integrating gives the CES demand equation in (21).

C3. Complete markets

The current value Lagrangian of the planner's problem is

$$\mathcal{L} = \sum_j \lambda_j u(C_j) + \lambda_j^{\text{RC}} \left(\left(\sum_{i=1}^N q_{ij}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} - C_j \right) + \lambda_j^{\text{GRC}} \left(\sum_{j=1}^N \tau_{jk} q_{jk} - z_j l_{Tj} \right) \quad (\text{C.5})$$

The first-order conditions are

$$C_j : \lambda_j u'(C_j) = \lambda_j^{\text{RC}} \quad (\text{C.6})$$

$$q_{ij} : \lambda_i^{\text{GRC}} \tau_{ij} = \lambda_j^{\text{RC}} C_j^{\frac{1}{\eta}} q_{ij}^{-\frac{1}{\eta}} \quad (\text{C.7})$$

Comparing these conditions to autarky's, it is evident under the normalization $\lambda_j^{\text{BC}} = 1/\lambda_j$ that the equilibrium prices of j 's tradable good, p_{jj} , are equal to the shadow values of j 's global resource constraint, λ_j^{GRC} .

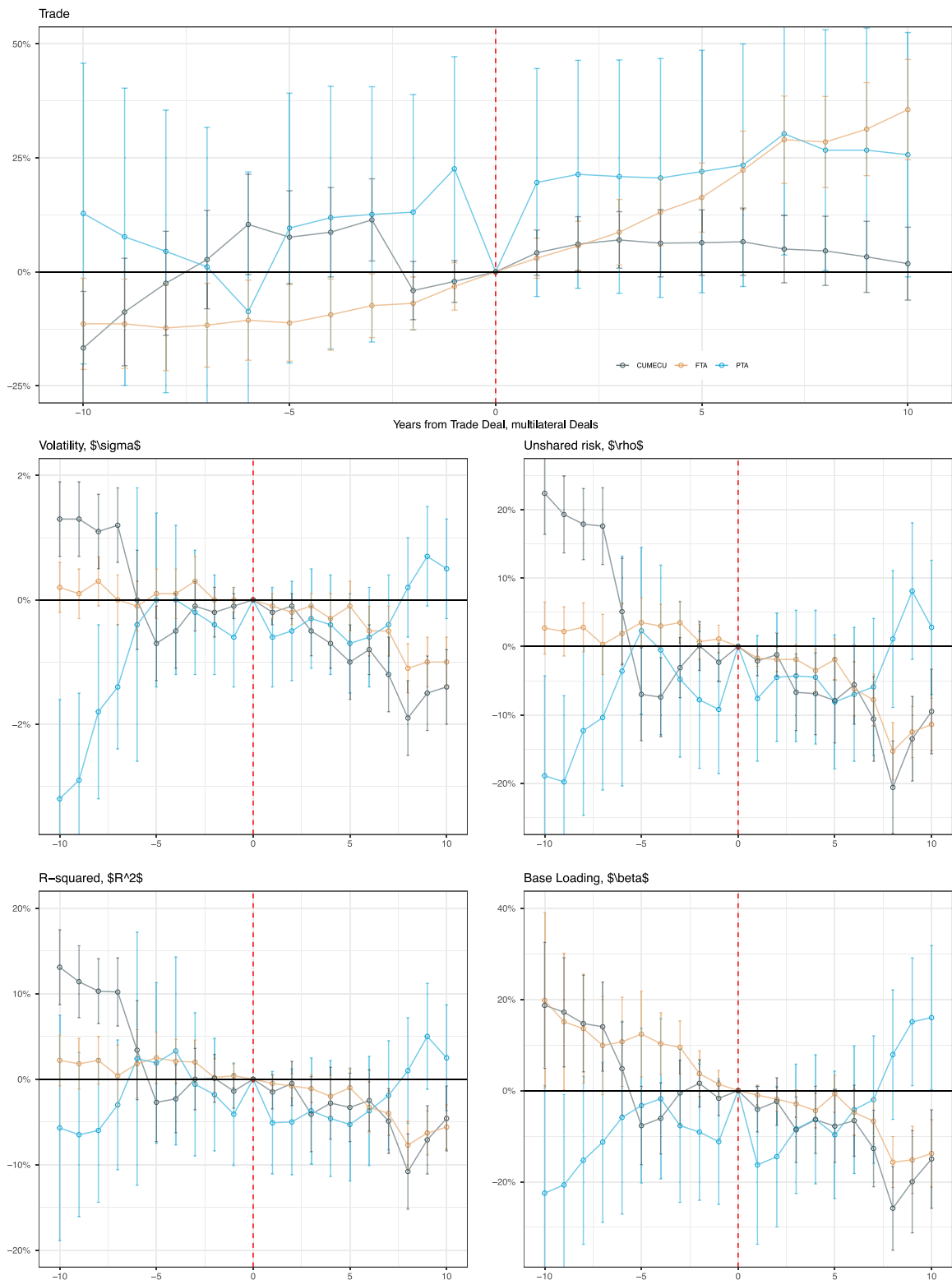


Fig. A.1. Change in Log Imports and Risk Around Trade Agreements By Deal Type. Multilateral Deals This figure plots the estimates of our staggered difference-in-difference regression in (12) with trade as the dependent variable. We compare for each year around a trade deal country-pairs with and without an agreement. The regression includes domestic and foreign country-year fixed effects and the standard gravity controls described in Table 2. Our trade deal variable equals one if countries have a multilateral FTA (yellow line), a multilateral CUMECU (gray line) or a multilateral PTA (blue line) and is zero otherwise.

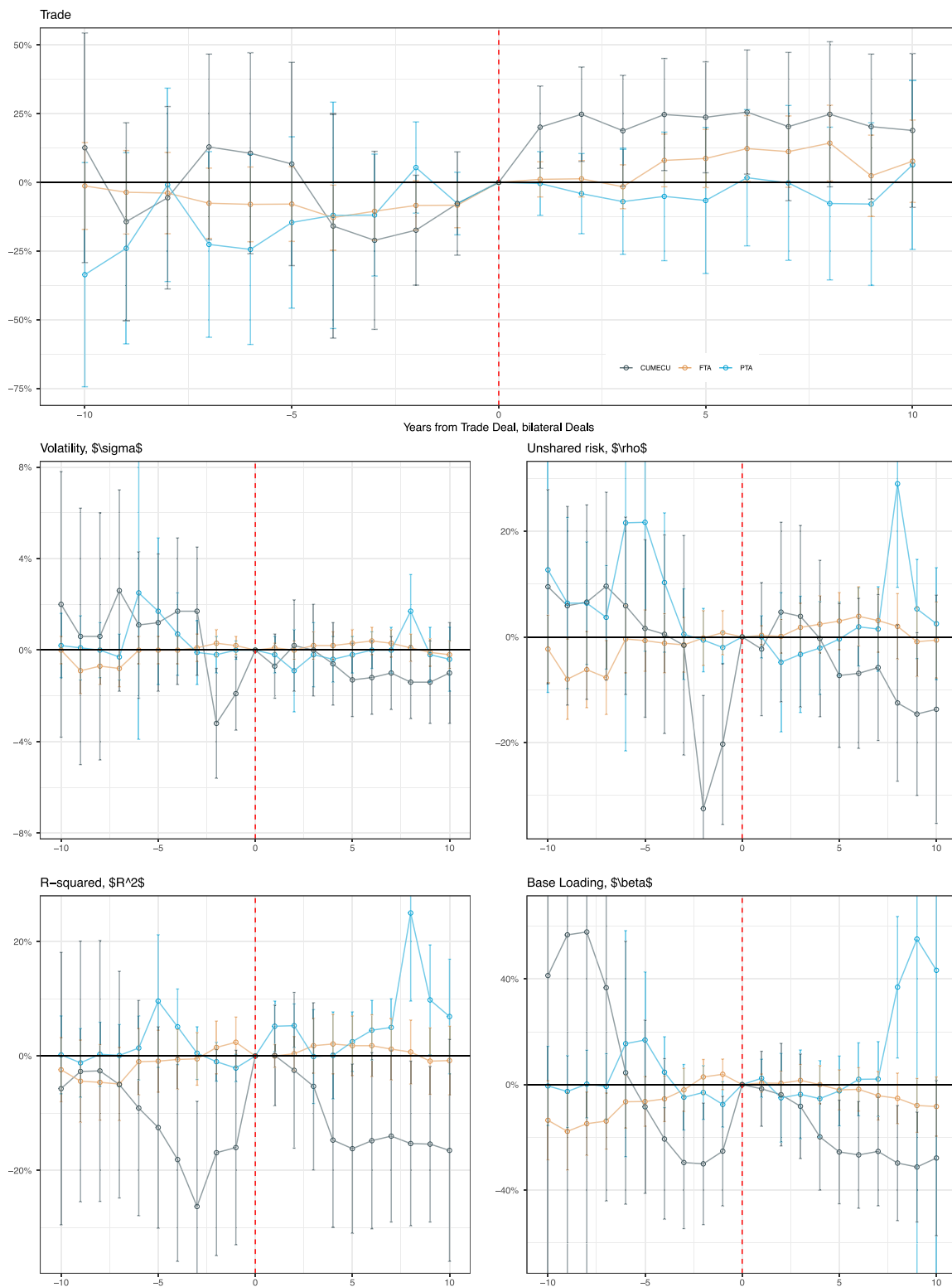


Fig. A.2. Change in Log Imports and Risk Around Trade Agreements By Deal Type. Bilateral Deals. This figure plots the estimates of our staggered difference-in-difference regression in (12) with trade as the dependent variable. We compare for each year around a trade deal country-pairs with and without an agreement. The regression includes domestic and foreign country-year fixed effects and the standard gravity controls described in Table 2. Our trade deal variable equals one if countries have a bilateral FTA (yellow line), a bilateral CUMECU (gray line) or a bilateral PTA (blue line) and is zero otherwise.

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